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Optimal plant layout considering the safety instrumented system design for hazardous equipment

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ABSTRACT

The safety of the process plant depends on the adequate separation between the assets and the hazardous units, along with the installation of protection devices. Several mathematical approaches have been proposed to reduce the risk of the explosions through the solution of the facility layout problem, but no model has included the design of the safety instrument systems. In this work, a MINLP approach was developed to solve three issues at the same time: the process equipment layout, the facility layout, and the safety instrumented system design. This approach aims to find the optimal facility layout that minimizes the land cost, the pipeline cost, and the lifecycle cost of each safety instrumented system, reducing the risk of explosions and keeping safety as much as possible the plant assets. The model was applied to find the optimal facility layout of ethylene oxide plant for different tolerable risk frequencies. In this way, this approach provides valuable information during the design stage and substantial support for decision-makers.

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1. Introduction

Plant layout design is a multidisciplinary task that requires the collaboration of different specialists. The objective is to provide the best arrangement of its elements, because it is a crucial factor in the economics and safety issues of the plant (Mannan, 2005). The design should minimize the amount of occupied land and piping used for the operation of the plant, guarantee access to individual units, and provide safety zones. The appropriate design will consider future expansions, environmental impact, efficiency and safety in the operations of the plant, the available area and operating expenses (Mecklenburgh, 1985). There are reviews of the approaches, solution methods, and formulations that have been proposed to find the optimal solution of the Facility Layout Problem (Anjos and Vieira, 2017; Drira et al., 2007; Singh and Sharma, 2006). Layout problem is commonly found in all kind of industries such as automotive, plastic, chemical, petrochemical, service, and others. Recent publications about this topic have focused on the optimization based on manufacturing cells system considering processes without uncertainty (Kumar and Singh, 2017, 2018a) and the probability of a sudden change in the process (Kumar and Singh, 2018b). These proposals can undoubtedly contribute significantly to processes with variable manufacturing. In spite of these investigations have focused on problems of minimizing the operating cost of the plant, the process safety is not considered; mainly because the hazards considered are noise, dangerous movements, and others concerning personal safety.

Regarding process safety, there are several efforts to reduce the risk by minimizing the consequences and frequency of accidents. Mathematical models were used to optimize the process equipment layout problem (PLP) and the facility layout problem (FLP). For PLP, Penteado and Ciric (1996) proposed a Mixed Integer Nonlinear Programming (MINLP) approach that includes the optimization of the selection of protection devices and the location of each equipment for ethylene oxide process plant; they assumed that the severity of the accident is 800% of the purchase cost of the affected equipment. Moreover, Patsiatzis et al. (2004) presented a Mixed Integer Linear

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Programming (MILP) model to determinate the coordinates and orientation of each equipment item, along with the type of protection devices; the severity of the accident was evaluated using the Dow Fire and Explosion Index system. Lopéz-Molina et al. (2013) used the multi-energy method to evaluate the overpressure to optimize the PLP and reduce the domino effect. The authors used the Domino Hazard Index to reduce the domino effect probability using a MINLP formulation (de Lira-Flores et al., 2014). Later, Latifi et al. (2017) proposed a MINLP formulation considering the toxic release, fire, and explosion scenarios using the Bat metaheuristic algorithm to solve the layout of a refinery unit. Recently, de Lira-Flores et al. (2018) proposed a MILP model based on quantitative risk analysis to reduce the risk through the layout optimization of storage tanks.

For the facility layout problem, Vázquez-Román et al. (2010) presented a MINLP considering the uncertainty of toxic release in one of the installed facilities; their formulation incorporates the effect of wind speed, wind direction, and atmospheric stability to calculate the risk of death via probit functions and Monte Carlo simulation. Moreover, Díaz-Ovalle et al. (2010) used a MINLP formulation to determine the optimal distribution of process facilities, taking into account the possibility of toxic release and the worst-case scenario, assuming deterministic values for wind and atmospheric conditions. In the same year, Jung et al. (2010a) used MILP approach for facility siting using mapping risks on a plant grid area. Later, Jung et al. (2010b) used a MINLP model for facility layout and siting, and the 3D-computational fluid dynamics (CFD) to compare the initial and final layouts for toxic release scenario. Also, Jung et al. (2011) used a MINLP formulation to risk reduction of damage buildings with occupants finding safe locations; in this case, the initial and final layouts were evaluated using FLACS. Rahman et al. (2014) solved an optimization problem using a MINLP approach to find the optimum locations of different facilities within the ammonia plant; they modeled the consequences due to toxic release and blast overpressure for various worst-case scenarios of the critical units. Also, Medina-Herrera et al. (2014) propose a MINLP model for the optimization of plant layout using quantitative risk analysis and bowtie analysis to identify possible catastrophic outcomes. Martinez-Gomez et al. (2015) proposed a MILP formulation for the optimal facility sitting and reallocation for future expansions, based on a multi-annual framework. Finally, Jung (2016) did a review of the current methodologies for plant layout optimization with safety considerations.

Moreover, pipe cost, land cost, risk cost, and protection devices cost are economic factors in most of the mathematical models presented in the literature. The pipe cost includes the capital investment and the operation cost of the total piping length; piping costs can be as high as 80% of the purchased equipment for several industries (Peters et al., 2003). On the other hand, land cost depends on the total occupied land area; this is a function of the safer founded distance between critical plant elements to minimize the consequences. Risk cost considers the losses assets, and represents the dollar value of expected losses caused by accidental events (fires or explosions). Protection cost depends on the safety devices installed at each unit to reduce the consequences or the frequency of the accidents. The accident consequences can be mitigated using passive safeguards and increasing the safety distance, but the land cost area and the pipe cost also increase when the gap is more significant. Protection devices are used to reduce the accident frequency or mitigate the accident severity and to avoid impractical distances during the operation process. For example, Penteado and Ciric (1996), and Patsiatzis et al. (2004) used the same protection devices for the ethylene oxide process. They used additional cooling water system, equipment fitted with other overpressure relief devices, process vessel with additional fire relief devices, a second skin on the reactor to protect against pressure, explosion protection system for the reactor, duplicated control shutdown system on absorption towers and duplicated control system with interlocking flow control on the reactor. The shutdown systems and the cooling water system are safety instrumented systems (SISs) that realize a safety instrumented function (SIF). However, the SIS design is an optimization problem by itself, which it was not included in the plant layout optimization models yet. Thus, the purpose of this work is to optimize the selection of the proper SIS elements for the dangerous equipment, while the plant layout design problem is solved. The most important considerations for SIS design are presented in the following paragraphs.

The IEC 61508 norm described the general SIS design (IEC, 2010; Smith and Simpson, 2016), while IEC 61511 is a technical standard norm for SIS implemented in process industries (Bond, 2002). The elements, for SIS design, taking into account are:

- Specific architecture (redundancy scheme).
- Hardware types: A or B.
- Multiple failure modes (safe and dangerous failure rates).
- Diagnostic coverage (share of detected and undetected failures).
- Common cause failure (β factor).
- Proof test intervals and repair times.
- The fault tolerance (FT) that is the number of faults that a subsystem can tolerate before resulting in system failure.
- The safe failure fraction (SFF) that is the percentage of subsystem failures considered safe.

The purpose of IEC61508 and IEC 61511 norms is the design of the SIS that achieve the Safety Integrity Level (SIL) appropriated. The SIL is a measure of the safety system performance required to control the risks to an acceptable level. A SIS consists of three categories of subsystems: sensors/transmitters, controllers, and final elements. These components work together to detect and prevent, or mitigate the effects of a hazardous event. The life cycle cost (LCC) for the SIS includes procurement, operation and risk costs. Goble (2010) presented an LCC model for SIS that was simplified by Torres-Echeverría et al. (2009), which was taken as a base on this proposal for SIS design. The LCC reduce the over or under design SIS, that is resulting in more expenses or operating with excessive risk.

An important remark regarding the risk accidents minimization by the solution of the facility and process equipment layout problems is that the research developed to find out the final design (through NLP, MILP, or MINLP approaches) have solved the problems separately but both are part of the same plant layout problem. Also, the safety instrumented system design problem has not been included during the layout optimization where it is a critical factor to determinate the safety of the process plant, in order to guarantee the tolerable risk criteria. For these reasons, a mathematical model to minimize the risk of accidents solving these problems in the same optimization process is required to improve the analysis of the decision-makers in the early stages of the plant design.

Thus, this paper presents a MINLP approach to minimize the total plant layout cost through the solution of the facility layout, process equipment layout and the safety instrumented systems design problems. The objective function consists of the total annual cost of the SIS, the total pipe cost, the land cost, and the financial risk. The risk was evaluated using QRA for explosion scenarios, while the case study is an ethylene oxide plant. The current approach can be used in a very high-risk process plant to optimize capital expenditures and operating expenses. The outline of the paper is as follows: Section 2 contains the description of the problem, Section 3 shows the mathematical

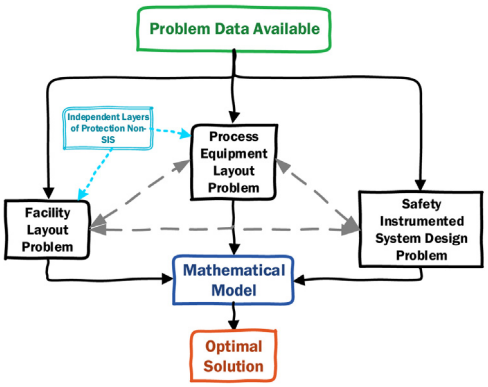


Fig. 1. Graphical representation of the problem statement.

model, while in Section 4 a case study is described. The discussion of the results is presented in Section 5. The concluding remarks of the work are outlined in Section 6.

2. Problem statement

The objective of this work is to find the optimal plant layout considering the facility layout, process equipment layout and the safety instrumented systems design problems. Fig. 1 shows a diagram to represent the problem to solve. The proposed model provides the solution of three problems, which are not independent: facility layout, process equipment layout, and safety instrumented system design for each hazardous equipment. The best arrangement of the process equipment and other facilities minimize the operation cost (defined by the pipe cost) and the land cost. The minimization of these costs produces a short separation distance between the units. However, the installation of hazardous units (tanks or equipment) could require considerable safety distance (increasing pipe and land cost).

For this reason, active and passive strategies are implemented in order to reduce the frequency or/and the consequences (for risk reduction) and to minimize the land area and pipe costs. For example, the frequency could be reduced with the implementation of standard operating procedures, basic process control systems, alarms with defined operator response, and safety instrumented systems (SIS); and the consequences are mitigated with the installation of pressure relief devices, blast walls and dikes, fire and gas systems, and deluge systems (Summers, 2003). In this work, safety instrumented systems design is achieved during the application of the mathematical model to satisfy the safety requirements of the plant. Therefore, the Safety Integrity Level depends on the separation distance between the hazardous equipment and the assets of the plan, if the probability of damage is significant for a short distance, then high SIL is required. Therefore, the three problems are interconnected by the separation distance, to get the best allocation of each equipment and facilities, Fig. 1. Also, according to Haitao and Xianhui (2009) the modeling of a SIS consuming a high amount of time, and altering a SIS design may lead to recreating the model; so it is difficult to compare one design with others effectively and quickly. Therefore, the optimization of the SIS is a significant problem. Other non-SIS independent layers of protections are not considered in the case study of this work, but these can be added before the optimization by reducing the initial failure rate or probability of damage, and its cost can be included in the cost of the equipment. Finally, the layout of industrial plants as addressed in this paper can be stated as follows.

Given

For Facility layout design

- A set of process equipment and facilities N, indexed i, j and k. NE is a set of process equipment units, $NE \subseteq N$, indexed ie, je and ke; and NF is a set of facilities, $NF \subseteq N$, such as process facilities, offices, parking lots, buildings, warehouses, storage tanks, utility areas, indexed if, jf and kf.
- A set of hazardous units $H \subseteq N$.
- A set of interconnected units *Inter*.
- Available land for siting facilities.
- The purchase and installation cost of piping, $CP_{i,j}$, between connected units in \$/m.
- The land cost per m^2.
- The initial failure rate per equipment or storage tank.
- Fire and explosion scenarios Data.
- Minimum safety distance between units, which are considered a source of hazards.
- Dangerous equipment and vessels

For SIS design:

- A set of technologies available *TEC*, (Hardware type: Type A or B)
- A set of subsystems *SUB*
- A set of specific architectures *RED* (redundancy scheme).
- Multiple failure modes (safe and dangerous failure rates)
- Diagnostic coverage (detected and undetected failures)
- Common Cause Failure (β factor)
- Proof Test Intervals TI
- Repair times TR.
- The Fault Tolerance (FT) (number of faults that a subsystem can tolerate before resulting in system failure)

- Safe Failure Fraction (SFF).

Determine:

- The allocation of each equipment or facility
- The optimal design of SIS for each dangerous equipment
- The minimum cost of the plant layout

To:

- Minimize the total plant layout cost

3. Mathematical formulation

The mathematical formulation proposes the minimization of an objective function that contains land, piping, financial risk, and safety instrumented system cost subject to the land, non-overlapping, facility orientation, available area, risk, and SIS lifecycle constraints. The model is described in detail in the next paragraphs.

3.1. Objective function

The objective function has three costs for the optimization of the plant layout: piping, land and the one associated with the SIS that combine the protection devices system and financial risk.

$$\min Cost_{Total} = \sum_{(i,j)\in Inter} CP_{i,j}\cdot D^{R}_{i,j} + LC\cdot AreaPlant + \sum_{i\in H} C^{SIS}_{i} \tag{1}$$

Where $CP_{i,j}$ pipe cost per \$/m of pipe length, *Inter* is the set of two interconnected equipment, $D^{R}_{i,j}$ is the rectilinear distance, *AreaPlant* is the area of the plant, and C^{SIS}_{i} is the SIS cost. This equation is subject to the distance, non-overlapping, facility orientation, plant area and lifecycle cost for SIS constraints, which are presented next.

3.2. Distance constraints

Euclidean, $D^{E}_{i,j}$, and Manhattan, $D^{R}_{i,j}$, distances are used for risk and pipe cost assessment, respectively. The Euclidean distance is used to evaluate the separation between two units from center to center, and its equation is:

$$D^{E}_{i,j} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \quad \forall\, i \in H, j \in N \tag{2}$$

where (x_i, y_i) are the center coordinates of the dangerous equipment, and $\left(x_j, y_j\right)$ are the center coordinates of process equipment or facility with the potential to be affected for an accident in $i-$ equipment. $D^{E}_{i,j}$ is essential to assess the damage effects produced by fire and explosion. On the other hand, the Manhattan distance is used to evaluate the pipe cost, and its equation is:

$$D^{R}_{i,j} = \left|x_i - x_j\right| + \left|y_i - y_j\right| \quad \forall i,j \in N,\ i \neq j;\ (i,j) \in Inter \tag{3}$$

This equation takes the sum of the absolute values of the differences of the coordinate. However, not all optimization software accepts absolute values; then, Eq. (3) is replaced with the following formulation:

$$D^{R}_{i,j} = Right_{i,j} + Left_{i,j} + Above_{i,j} + Below_{i,j} \quad \forall i \neq j;\ i,j \in N \tag{4}$$

$$x_i - x_j = Right_{i,j} - Left_{i,j} \quad \forall i \neq j;\ i,j \in N \tag{5}$$

$$y_i - y_j = Above_{i,j} - Below_{i,j} \quad \forall i \neq j;\ i,j \in N \tag{6}$$

$$Right_{i,j} \leq LPLANT_x \cdot Z^{x}_{i,j} \quad \forall i \neq j;\ i,j \in N \tag{7}$$

$$Left_{i,j} \leq LPLANT_x \left(1 - Z^{x}_{i,j}\right) \quad \forall i \neq j;\ i,j \in N \tag{8}$$

$$Above_{i,j} \leq LPLANT_y \cdot Z^{y}_{i,j} \quad \forall i \neq j;\ i,j \in N \tag{9}$$

$$Below_{i,j} \leq LPLANT_y \left(1 - Z^{y}_{i,j}\right) \quad \forall i \neq j;\ i,j \in N \tag{10}$$

where: $LPLANT_x$ and $LPLANT_y$ are the length and width of the available area respectively; $Z^{x}_{i,j}$ and $Z^{y}_{i,j}$ are 0–1 binary variables; $Right_{i,j}$, $Left_{i,j}$, $Above_{i,j}$ and $Below_{i,j}$ are non-negative slack variables used to know the position of the one $i-$ unit respect $j-$ unit. The sub-indexes *if* and *jf* or *ie* and *je* are substituted instead of *i* and *j* for facilities or equipment respectively.

3.3. Non-overlapping constraints

The non-overlapping equations are taking from Meller et al. (2007). The equations below are known as sequence-pair formulation. The Eqs. (11)–(14) improve the convergence to find a feasible solution.

$$Z^{1}_{i,j} + Z^{1}_{j,i} = 1 \quad \forall i,j \in N,\ \ i \neq j \tag{11}$$

$$Z^{2}_{i,j} + Z^{2}_{j,i} = 1 \quad \forall i,j \in N,\ \ i \neq j \tag{12}$$

$$Z^1_{i,k} \geq Z^1_{i,j} + Z^1_{j,k} - 1 \quad \forall i,j,k \in N, \;\; i \neq j \neq k \tag{13}$$

$$Z^2_{i,k} \geq Z^2_{i,j} + Z^2_{j,k} - 1 \quad \forall i,j,k \in N, \;\; i \neq j \neq k \tag{14}$$

Where $Z^1_{i,j}$ and $Z^2_{i,j}$ are binary variables [0–1]. Also, these equations can be applied to both facility layout and process equipment layout problems. However, non-overlapping constraints have little differences for both problems. The non-overlapping constraints for facility layout problem are:

$$x_{if} - x_{jf} + \frac{l^x_{if} + l^x_{jf}}{2} + STF \leq LPLANT_x \left(2 - Z^1_{if,jf} - Z^2_{if,jf}\right) \quad \forall if, jf \in NF, \;\; if \neq jf \tag{15}$$

$$y_{if} - y_{jf} + \frac{l^y_{if} + l^y_{jf}}{2} + STF \leq LPLANT_y \left(1 + Z^1_{if,jf} - Z^2_{if,jf}\right) \quad \forall if, jf \in NF, \;\; if \neq jf \tag{16}$$

where $LPLANT_x$ and $LPLANT_y$ are the length and width dimensions of the total available plant area, respectively. $Z^1_{if,jf}$, and $Z^2_{if,jf}$ are $Z^1_{i,j}$ and $Z^2_{i,j}$ for facilities. *STF* is a street distance between two facilities. l^x_{if} and l^y_{if} are the length and width dimensions of the *if* -facility and depends on its orientation. The non-overlapping constraints for process equipment layout are:

$$x_{ie} - x_{je} + \frac{l^x_{ie} + l^x_{je}}{2} + STE \leq LP_x \left(2 - Z^1_{ie,je} - Z^2_{ie,je}\right) \quad \forall ie, je \in NE, \;\; ie \neq je \tag{17}$$

$$y_{ie} - y_{je} + \frac{l^y_{ie} + l^y_{je}}{2} + STE \leq LP_y \left(1 + Z^1_{ie,je} - Z^2_{ie,je}\right) \quad \forall ie, je \in NE, \;\; ie \neq je \tag{18}$$

where LP_x and LP_y the length and width dimension of the process facility area, respectively. $Z^1_{ie,je}$ and $Z^2_{ie,je}$ are $Z^1_{i,j}$ and $Z^2_{i,j}$ for process equipment. *STE* parameter is the minimum separation distance between two equipment required for access to the equipment for maintenance. l^x_{ie} and l^y_{ie} are the length and width dimensions of the equipment.

3.4. Facility orientation constraints

The orientation equations for a facility with a rectangular shape are:

$$l^x_{if} = DimFac1_{if} \cdot B^o_{if} + DimFac2_{if} \cdot \left(1 - B^o_{if}\right) \quad \forall if \in NF - \left\{Process\right\} \tag{19}$$

$$l^y_{if} = DimFac1_{if} \cdot B^o_{if} + DimFac2_{if} \cdot \left(1 - B^o_{if}\right) \quad \forall if \in NF - \left\{Process\right\} \tag{20}$$

where: $DimFac1_{if}$ and $DimFac2_{if}$ are the dimensions of the $if-$ facility, and B^o_{if} is 0–1 binary variable. However, the process area dimensions are a result of the optimization process, because all process equipment must be inside the processing facility; but the process equipment locations are variables. The process dimensions will be evaluated with the following equations:

$$l^x_{Process} = l^{x,UP}_{Process} - l^{x,LO}_{Process} \tag{21}$$

$$l^y_{Process} = l^{y,UP}_{Process} - l^{y,LO}_{Process} \tag{22}$$

$$l^{x,UP}_{Process} \geq x_{ie} + \frac{l^x_{ie}}{2} + STE \quad \forall ie \in NE \tag{23}$$

$$l^{x,LO}_{Process} \leq x_{ie} - \frac{l^x_{ie}}{2} - STE \quad \forall ie \in NE \tag{24}$$

$$l^{y,UP}_{Process} \geq y_{ie} + \frac{l^y_{ie}}{2} + STE \quad \forall ie \in NE \tag{25}$$

$$l^{y,LO}_{Process} \leq y_{ie} - \frac{l^y_{ie}}{2} - STE \quad \forall ie \in NE \tag{26}$$

$$x_{ie} + \frac{l^x_{ie}}{2} + STE \leq x_{Process} + \frac{l^x_{Process}}{2} \quad \forall ie \in NE \tag{27}$$

$$x_{ie} - \frac{l^x_{ie}}{2} - STE \geq x_{Process} - \frac{l^x_{Process}}{2} \quad \forall ie \in NE \tag{28}$$

$$y_{ie} + \frac{l^y_{ie}}{2} + STE \leq y_{Process} + \frac{l^y_{Process}}{2} \quad \forall ie \in NE \tag{29}$$

$$y_{ie} - \frac{l^y_{ie}}{2} - STE \geq y_{Process} - \frac{l^y_{Process}}{2} \quad \forall ie \in NE \tag{30}$$

$$x_{ie} \geq \frac{l^x_{ie}}{2} + STE \quad \forall ie \in NE \tag{31}$$

$$y_{ie} \geq \frac{l^y_{ie}}{2} + STE \quad \forall ie \in NE \tag{32}$$

$$x_{ie} \leq l^{x}_{Process} - \frac{l^{x}_{ie}}{2} - STE \quad \forall ie \in NE \tag{33}$$

$$y_{ie} \leq l^{y}_{Process} - \frac{l^{y}_{ie}}{2} - STE \quad \forall ie \in NE \tag{34}$$

Eqs. (19) and (20) define the length and width of the process area. Eqs. (21)–(34) guarantee that each equipment is located inside the process area.

3.5. Plant area constraints

The land cost depends on the total area occupied, *AreaPlant*. The plant area is evaluated using the following equations proposed by Vázquez-Román et al. (2010):

$$AreaPlant = LPlant \cdot DPlant \tag{35}$$

$$LPlant \geq x_{if} + \frac{l^{x}_{if}}{2} \quad \forall if \in NF \tag{36}$$

$$DPlant \geq y_{if} + \frac{l^{y}_{if}}{2} \quad \forall if \in NF \tag{37}$$

$$0 \leq LPlant \leq LPLANT_{x} \tag{38}$$

$$0 \leq DPlant \leq LPLANT_{y} \tag{39}$$

$$\frac{l^{x}_{if}}{2} + STF \leq x_{if} \leq LPLANT_{x} - \frac{l^{x}_{if}}{2} - STF \quad \forall if \in NF \tag{40}$$

$$\frac{l^{y}_{if}}{2} + STF \leq y_{if} \leq LPLANT_{y} - \frac{l^{y}_{if}}{2} - STF \quad \forall if \in NF \tag{41}$$

where *LPlant* and *DPlant* are length and width dimensions of the total occupied plant area, respectively.

3.6. Damage cost constraints

The damage probability value, $P_{Di,j}$, is evaluated using the mathematical formulation proposed by López-Molina et al. (2013):

$$P_{D}i_{,j} = \frac{1.005}{1 + e^{-\left(\frac{Y_{i,j} - 5.004}{0.6120}\right)}} \quad \forall i,j \in N; i \in H; i \neq j \tag{42}$$

$$\Delta P_{i,j} = 134.1196 \cdot e^{-0.8967 \bullet Ro} \left(1 - \propto_{i,j}\right) + 32.946 \cdot Ro_{i,j}^{-1.121} \cdot \propto \quad \forall i,j \in N; i \in H; i \neq j \tag{43}$$

$$Ro_{i,j} = \left(\frac{E_i}{P_a}\right)^{-1/3} \cdot D^{E}_{i,j} \quad \forall i,j \in N; i \in H; i \neq j \tag{44}$$

$$\propto_{i,j} = \frac{Ro_{i,j} - 2}{2 \cdot \sqrt{1 + \left(Ro_{i,j} - 2\right)^{2}}} + \frac{1}{2} \quad \forall i,j \in N; i \in H; i \neq j \tag{45}$$

where: $\propto_{i,j}$ is a smoothing factor. $Ro_{i,j}$ is the characteristic distance for the multi-energy method. E_i is the energy explosion. *Pa* is the atmospheric pressure. $\Delta P_{i,j}$ is the overpressure peak produced by an explosion in the equipment *i* over the equipment *j*. $Y_{i,j}$ is the probit value, which depends on the potential target of the explosion. There are several models to evaluate the consequences, e.g., Khan et al. (2015) provide a review of methods and models for risk assessment, including probit models. The probit equations are used to evaluate the damage of specific targets, i.e., there are models to estimate the probability of injury or likelihood of death due to thermal radiation, toxic gas, and blast overpressure effects. Other probit models predict the damage probability of atmospheric tanks, pressurized vessels or process equipment by fire or explosion. In this work, the used probit models quantify the probability of accident propagation (domino effect) to avoid catastrophic events. Under this consideration, only the process plant elements such as tanks, vessels, towers, and reactors have the potential to produce a domino effect as a consequence of an initial hazardous event. The probit models suitable for domino effect were proposed by Cozzani et al. (2005) and Mingguang and Juncheng (2008).

On the other hand, Eisenberg et al. (1975) presented a probit to estimate the structural damage for buildings and several probit models to quantify the damage over people by fire and explosion. Also, Días-Alonso et al. (2008) provide equations to evaluate the probability of reinforced concrete structures (building and factory) failure caused by vapor cloud explosions. An analysis of which model is better for each different situation was not made, since the focus of this research was the applicability of the MINLP approach. Therefore, the cited models can be used to predict the damage probabilities for explosions because there is not enough evidence to determine which model is the best for all cases, e.g., when a study requires more conservatives results. The models from Mingguang and Juncheng (2008) and Eisenberg et al. (1975) were selected to evaluate $Y_{i,j}$. However, the models by Cozzani et al. (2005) and Días-Alonso et al. (2008) can be used too. Then, the equations are:

- For the set *NE* of process equipment (Mingguang and Juncheng, 2008):

$$Y_{i,j} = -12.22 + 1.65 \ln\left(\Delta P_{i,j}\right) \quad \forall i \in H; j \in NE; i \neq j \tag{46}$$

- For the set *A* of atmospheric tanks (Mingguang and Juncheng, 2008):

$$Y_{i,j} = -9.36 + 1.43 \ln\left(\Delta P_{i,j}\right) \quad \forall i \in H; j \in A; i \neq j \tag{47}$$

- For the set *P* of a pressurized vessel (Mingguang and Juncheng, 2008):

$$Y_{i,j} = -14.44 + 1.82 \ln\left(\Delta P_{i,j}\right) \quad \forall i \in H; j \in P; i \neq j \tag{48}$$

- For the set *B* of buildings is used (Eisenberg et al., 1975):

$$Y_{i,j} = -23.8 + 2.92 \ln\left(\Delta P_{i,j}\right) \quad \forall i \in H; j \in B; \tag{49}$$

In this work, $NF = A \cup P \cup B \cup \{Process\}$. The risk cost, C_i^{Loss}, produced by the hazardous unit is calculated using the equation used by Tugnoli et al. (2008):

$$C_i^{Loss} = C_i^{PAHE} + C_i^{CDE} \quad \forall i \in H \tag{50}$$

where: C_i^{PAHE} combines several costs such as C_i^{PL} is the production loss; C_i^{AL} is the purchase value of the unit; C_i^{HHL} is the human health loss calculated in terms of the cost of fatalities/injuries employing insurance costs, rehabilitation costs, and worker compensation rates directly caused by accident at the unit (HSE); but it is important to clarify that all human life is immeasurable. C_i^{ECC} and is associated with the mass or volume of soil, water and air that were contaminated by the hazardous accident; and C_i^{CDE} represents the sum of the loss related to the secondary units involved:

$$C_i^{CDE} = \sum_j P_D i_{,j} \left(C_j^{PL} + C_j^{AL} + C_j^{HHL} + C_j^{ECC}\right) \quad \forall i \in H \tag{51}$$

Then, both Eqs. (50) and (51) are combined to represent the catastrophic cost of the accident. Gavious et al. (2009) proposed a model related with costs involved in industrial accidents. They divided the accident cost in direct, indirect, payment, and immeasurable costs. These cost are included into the C_i^{PAHE} cost calculation. The major contribution of Gavious et al. (2009) was to identify the importance of evaluating indirect costs (the cost per recruitment, time spent on training and the generation of delay in the production and bottlenecks). However, they dont́ include the C_i^{ECC}. Therefore, the cost C_i^{Loss} is related with the propagation of the accident.

On the other hand, the damage probability value affects the safety of the units; for this reason, independent protection layers are installed in hazardous equipment to reduce the frequency of the accident. The SIS is one of the most critical and expensive layers, and its design requires to achieve an acceptable Safety Integrity Level. The LCC of SIS design model consists in the next formulation.

3.7. Lifecycle cost for SIS constraints

The model presented in this section is based on Goble (2010) and Torres-Echeverría et al. (2009), but it was adapted as a MINLP approach. The SIS cost is divided into initial fixed and annual costs:

$$C_i^{SIS} = C_i^{Fixed} + \left(C_i^{Op} + C_i^{Risk}\right) PVF \quad \forall i \in H \tag{52}$$

where: C_i^{Fixed} are the initial fixed costs such as the procurement costs, C^{DPIC} considers system design cost, purchase cost of the equipment including initial training, installation cost, and start-up/system commissioning cost, $C_i^{START-UP}$. These costs occur only once. On the other hand, the operating cost, C_i^{Op}, includes the ones associated to shut down, system engineering changes, energy consumption, parts consumption, and fixed and variable maintenance (Goble, 2010). The risk cost, C_i^{Risk}, is the failure cost. These costs are incurred each year. The present value annuity factor (*PVF*) is used to calculate the present value of the future one dollar cash flows for the operating and risk costs. The present value annuity factor is defined as:

$$PVF = \frac{1 - (1+R)^{-n}}{R} \tag{53}$$

where: *R* is the interest rate, and *n* is the number of periods.

3.7.1. Procurement costs constraints

The total procurement cost is obtained by summing all subsystem procurement costs:

$$C_i^{Fixed} = \sum_s C_{i,s}^{DPIC} + C_i^{START-UP} \quad \forall i \in H; \tag{54}$$

where the start-up cost is independent of subsystems cost, $C_{i,s}^{DPIC}$ which includes the procurement costs per subsystem $s \in SUB$. These costs are evaluated with the next equations:

$$C_{i,s}^{DPIC} = \sum_{tr} C_{i,s,tr}^{DPIC} \cdot B^{i,s,tr} \quad \forall i \in H; \forall s \in SUB \tag{55}$$

$$\sum_{tr} B^{i,s,tr} = 1 \quad \forall i \in H; \forall s \in SUB \tag{56}$$

where $C_{i,s,tr}^{DPIC}$ are the procurement costs per subsystem $s \in SUB$, and technology-architecture $tr \in TR$; $B_{i,s,tr}$ is a 0–1 binary variable, which is used to decide on the subsystem configuration. The TR set is the k-out-of-n different structure per subsystem, n is the number of redundancies. The redundancies could be with homogeneous or heterogeneous technology. Therefore, in the remainder of the document, the sub-indexes i, s, tr mean that the calculation of the variable is done per hazardous equipment, subsystem, and technology-architecture structure available, respectively.

3.7.2. Operating costs constraints

In this model, the SIS operating cost per hazardous equipment, C_i^{OP}, includes the costs of preventive maintenance (C_i^{PM}), testing (C_i^{T}) and corrective maintenance (C_i^{CM}). The equation is as follows:

$$C_i^{OP} = C_i^{T} + C_i^{PM} + C_i^{CM} \quad \forall i \in H \tag{57}$$

These costs are divided by subsystem. Then, the formulation is:

$$C_i^{T} = \sum_{s} C_{i,s}^{T} \quad \forall i \in H; \tag{58}$$

$$C_i^{PM} = \sum_{s} C_{i,s}^{PM} \quad \forall i \in H; \tag{59}$$

$$C_i^{CM} = \sum_{s} C_{i,s}^{CM} \quad \forall i \in H; \tag{60}$$

The decision of architecture and technology used for each subsystem is taking with:

$$C_{i,s}^{T} = \sum_{tr} C_{i,s,tr}^{T} \cdot B^{i,s,tr} \quad \forall i \in H; \forall s \in SUB \tag{61}$$

$$C_{i,s}^{PM} = \sum_{tr} C_{i,s,tr}^{PM} B^{i,s,tr} \quad \forall i \in H; \forall s \in SUB \tag{62}$$

$$C_{i,s}^{CM} = \sum_{tr} C_{i,s,tr}^{CM} \cdot B^{i,s,tr} \quad \forall i \in H; \forall s \in SUB \tag{63}$$

3.7.3. Risk cost constraints

The risk cost (C_i^{Risk}) consists of two costs: spurious trip cost (C_i^{STR}) and the hazard cost (C_i^{HAZ}).

$$C_i^{Risk} = C_i^{STR} + C_i^{HAZ} \quad \forall i \in H \tag{64}$$

The cost of the spurious trip rate C_i^{STR} is directly proportional to the cost of production loss from each spurious shutdown (CSD_i):

$$C_i^{STR} = STR^{i} \cdot CSD^{i} \quad \forall i \in H \tag{65}$$

$$CSD^{i} = SD_i^{Loss} \cdot SD_i^{Time} \quad \forall i \in H \tag{66}$$

$$STR^{i} = 8760 \sum_{s} STR^{i,s} \quad \forall i \in H; \tag{67}$$

$$STR^{i,s} = \sum_{tr} STR^{i,s,tr} B^{i,s,tr} \quad \forall i \in H; \forall s \in SUB \tag{68}$$

where: STR is the spurious trip rate; 8760 number is the total hours in a year because $STR_{i,s}$ has units of hours^{-1} (the same units of failure rate λ), and STR_i is required in units of years^{-1}; SD_i^{Loss} is the loss of production per hour, SD_i^{Time} is the plant restoration downtime after a spurious trip. On the other hand, the hazard cost is estimated with Eq. (69):

$$C_i^{HAZ} = C_i^{Loss} \cdot f_i^{I} \cdot PFD_i^{Av} \quad \forall i \in H \tag{69}$$

$$PFD_i^{Av} = PFD_i^{NON-SIS} \cdot PFD_i^{SIS} \quad \forall i \in H \tag{70}$$

where: C_i^{Loss} is evaluated using Eq. (50) and depends on the layout configuration, f_i^{I} is the initial event frequency of the accident, $PFD_i^{NON-SIS}$ is the probability of failure on demand of non-SIS independent protection layers, PFD_i^{SIS} is the probability of failure on demand of SIS, and PFD_i^{Av} is the probability of failure on demand average (the result of the PFD of each subsystem). However, the sensors can be of different types such as temperature, pressure and filling level. The PFD calculation for this subsystem is different. For example, a process has installed two sensors on each equipment, one for temperature and one for pressure. Both sensors are used to detect a deviation of the normal condition of the equipment to prevent an explosion. Then, the PFD_i^{SIS} is calculated with the following equation:

$$PFD_i^{SIS} = \left(PFD^{i,ST} \cdot PFD^{i,SP}\right) + PFD^{i,LS} + PFD^{i,FE} \quad \forall i \in H \tag{71}$$

Table 1
Units dimensions and C_i^{PAHE} cost (production loss, purchase, human health loss cost).

Unit	Symbol	Type	$Dim1_i$ (m)	$Dim2_i$ (m)	C_i^{PAHE} (USD thousand)
Process-Facility*	PF	Facility	–	–	–
Control-Room	CR	Facility	15	20	7,800.00
Effluent-Treatment-Plant	ETP	Facility	35	20	3,420.00
Ethylene Oxide Tanks	EOT	Facility	20	10	3,544.30
Ethylene Tanks	ET	Facility	10	10	1,210.70
Solvent Storage Tanks	SST	Facility	40	11	490.00
Storage Area	SA	Facility	40	27.5	3,415.00
Administration Buildings	AB	Facility	40	20	31,800.00
Laboratory	LAB	Facility	20	10	9,380.00
Reactor	1	Equipment	5.22	5.22	1,255.00
Heat Exchanger A	2	Equipment	11.42	11.42	45.00
Ethylene Oxide Absorber	3	Equipment	7.68	7.68	937.00
Heat Exchanger B	4	Equipment	8.48	8.48	38.00
CO_2 Absorber	5	Equipment	7.68	7.68	891.30
Flash Tank	6	Equipment	2.6	2.6	39.00
Pump	7	Equipment	2.4	2.4	9.50

* Length and depth of process facility are variables.

where $PFD_{i,ST}$, $PFD_{i,SP}$, $PFD_{i,LS}$, and $PFD_{i,FE}$ are the PFD of the temperature sensor, pressure sensors, logic solver, and final element, respectively. The $PFD_{i,s}$ of the subsystem is calculated using:

$$PFD^{i,s} = \sum_{tr} PFD^{i,s,tr} B^{i,s,tr} \quad PFD^{i,s} = \sum_{tr} PFD^{i,s,tr} B^{i,s,tr} \tag{72}$$

Also, the required PFD_i^{Av} value depends on the tolerable risk frequency (F_{TR}) or risk target per year; this value is fixed. The necessary risk reduction is equivalent to the PFD_i^{Av}, determined by the expression:

$$F_{TR} \geq f_i^I \cdot PFD_i^{Av} \quad \forall i \in H; \forall s \in SUB \tag{73}$$

Finally, the parameters $C_{i,s,tr}^{DPIC}$, $C_{i,s,tr}^{T}$, $C_{i,s,tr}^{PM}$, $C_{i,s,tr}^{CM}$, $STR_{i,s,tr}$, and $PFD_{i,s,tr}$ depend of:

- The set of technologies available (TEC)
- The set of specific architectures (RED)
- Multiple failure modes and diagnostic coverage: safe detected ($\lambda_{p,s,t}^{SD}$), safe undetected ($\lambda_{p,s,t}^{SU}$), dangerous detected ($\lambda_{p,s,t}^{DD}$), and dangerous undetected ($\lambda_{i,s,t}^{DU}$) failure rates.
- Common Cause Failure (factor). C_{MooN} is the configuration factor.
- Proof Test Intervals TI
- Repair times TR.
- The Fault Tolerance (FT) (number of faults that a subsystem can tolerate before resulting in system failure)

There are different techniques to assess $STR_{i,s,tr}$ and $PFD_{i,s,tr}$, such as Markov models Reliability Block Diagrams and Fault Tree Analysis. In this work, these variables are quantified based on Fault Tree Analysis. The Appendix shows the equations used during the solution of the study case, which is presented next.

4. Case study – ethylene oxide process plant

An ethylene oxide process plant is studied. The process includes two parallel exothermic reactions, which are carried out on the reactor:

$$C_2H_2 + 0.5O_2 \rightarrow C_2H_4O \tag{74}$$

$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O \tag{75}$$

Fig. 2 shows the process flow diagram and a graphical representation of the problem to solve. The interconnected equipment requires being close to reducing the operation cost. However, ethylene vessel, ethylene oxide vessel, reactor, and both absorbers are hazardous equipment, because explosions could be occurring by loss process control and release of hazardous material, such as ethylene or ethylene oxide. Therefore, the mathematical model is applied to optimize the SIS per hazardous equipment and the location of each unit (facilities and equipment).

The unit dimensions and C_i^{PAHE} costs are given in Table 1. The C_i^{PAHE} is the cost of unit i, without domino effect. The C_i^{PAHE} value was evaluated considering a five, two, two, twenty, and six workers in the control room, effluent-treatment plant, storage area, administration buildings, and laboratory, respectively. The human health loss cost was estimated using data provided by the Health and Safety Executive (HSE, 2015). In this case, the pair administration buildings are the most expensive facilities for a catastrophic scenario because the risk cost of injuries or fatalities is high. Therefore, the minimization of the total risk cost will provide a safe location of the critical units. The equipment cost and dimensions were taken from Patsiatzis et al. (2004). The land available for sitting the facilities has 400 m and 125 m for length and width respectively. The minimum separation space between two equipment (STE) is 2 m, and between two facilities (STF) is 0 m. The $CP_{i,j}$ is $ 83.4 /m, and the cost of land is $ 6.0 /m^2.

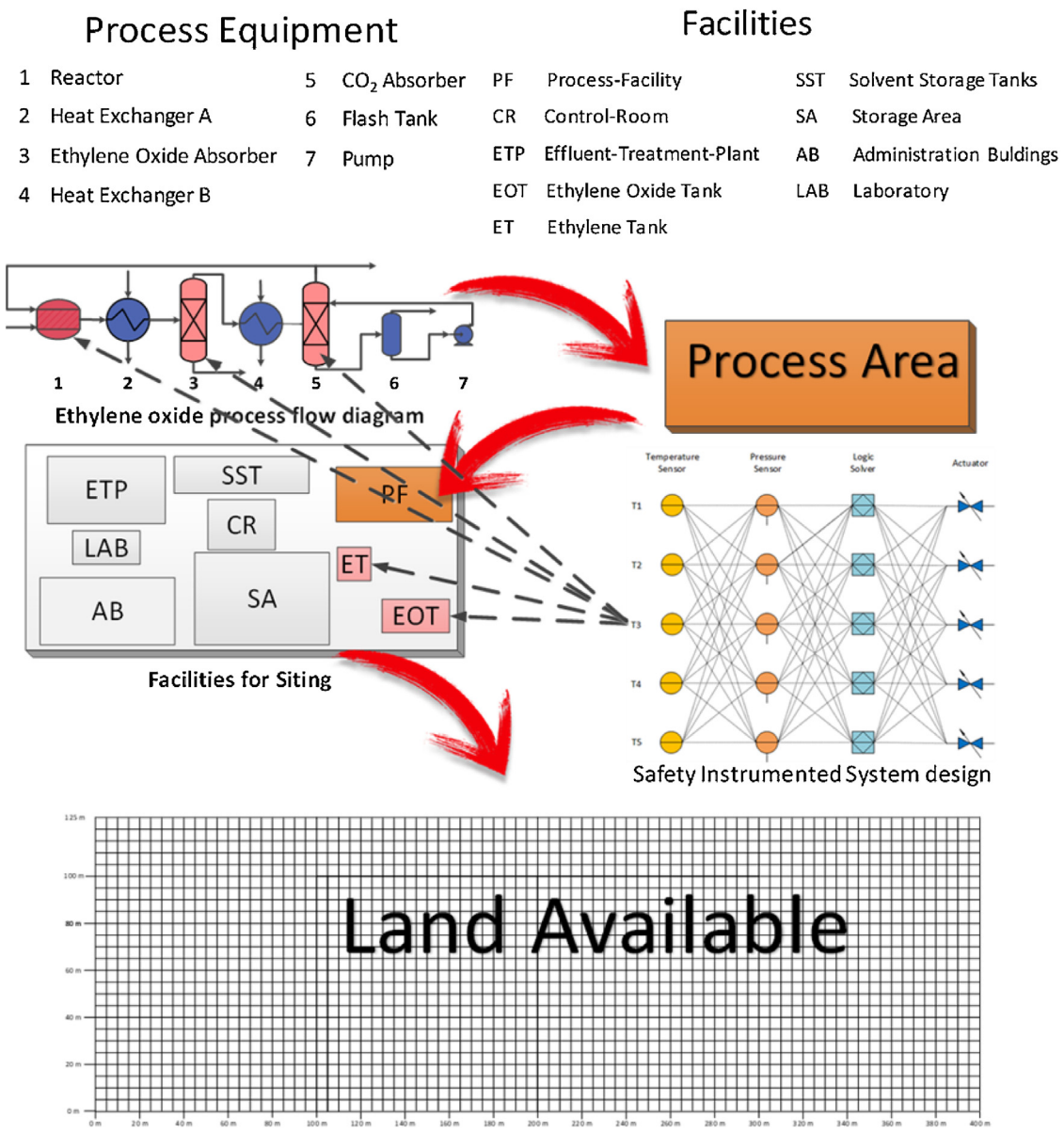


Fig. 2. Schematic representation of the addressed problem.

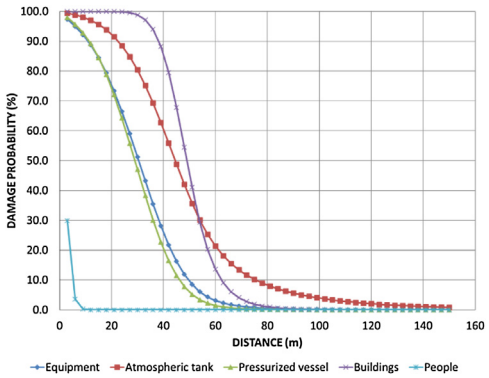


Fig. 3. Damage probability for ethylene oxide vessel explosion versus distance.

The $\left(E_i/P_a\right)^{1/3}$ distance values calculated for this problem were 10.66, 6.08, 5.75, 21.84, 27.34 m for the reactor, ethylene oxide absorber, CO_2 absorber, ethylene oxide vessel, and ethylene vessel, respectively. These values were estimated considering a pipeline rupture of 10″ diameter. Then, the damage probability is calculated as a function of the distance. For example, Fig. 3 shows the damage probability for

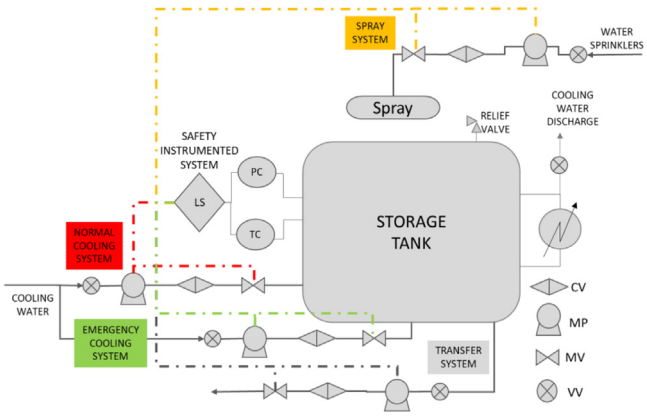


Fig. 4. EO Vessel safety system diagram base on the data presented by Martorell et al. (2010).

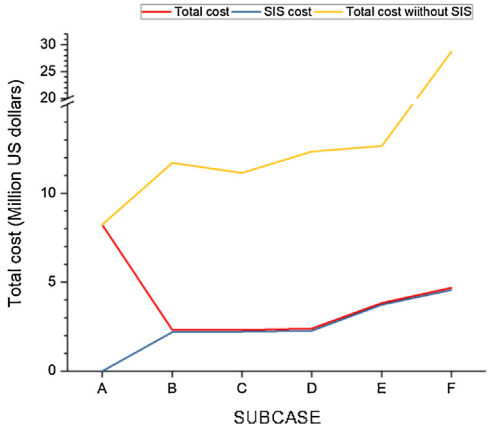


Fig. 5. Total cost comparison for the optimal solutions.

ethylene oxide vessel explosion for different targets, where the Eqs. (42)–(49) were used. For atmospheric tanks and the buildings, the damage probability is reduced around of the 80% in the first 60 m, and the same probability for small equipment and pressurized vessels but in the first 40 m. Besides, the principal hazard of injuries or fatalities for workers is related to the collapsed buildings, when they are inside (Fig. 4).

Active independent protection layers can be taken into account as measures to correcting the problem, risk reduction, in time such as the Basic Process Control System (BCPS), high-temperature alarm, both low and high-level pressure alarm, and the operator actions to reestablishing the coolant flow. For example, ethylene oxide vessel has four bars and 10 °C as storage conditions. All necessary safety systems are provided to the facility; these safety systems are an emergency cooling system, which acts after a failure of the cooling system. The transfer system moves the ethylene oxide from the vessel to an auxiliary tank. The spray system is responsible for cooling the vessel after a failure of the emergency cooling systems, given a failure of the transfer system, is produced. Relief valve system is composed of one safety-relief valve, and it provides the last safety measure to prevent a tank explosion due to an overpressure if the previous safety functions have failed. Valves open automatically to keep the pressure under control, although this action means there is a release of gas into the atmosphere. All these systems, except the last one, are controlled by a safety instrumented system. This system consists of temperature and pressure sensors, and transmitters. The last ones send the signal to logical solvers that are responsible for the actuation of the controlled valves and pumps. The SIS acts in case of a high temperature/pressure beyond the safety limits (Martorell et al., 2010). In this study case, for academic purposes, the same initial failure rate of the accident is used ($1 \times 10^{-1}\text{yr}^{-1}$ for all hazardous unit). The optimization of the safety distance and the SIS design are implemented to reduce the risk. The applications of additional layers of protections reduce the SIL requirements, which is an advantage, but the total amount of potential solutions of SIS design optimization problem is reduced.

The Safety Instrumented System design includes both temperature and pressure sensors, PLC and valve. Tables 2–6 show the failure rate per year and beta factor data per subsystem and technology. The multiple failure rates are included (safe detected, safe undetected, dangerous detected, and dangerous undetected). Tables 7–11 present the life cycle cost data for the hazardous units (the cost due to purchase, design, installation/consumption, preventive maintenance, repair per instrument, test interval maintenance) per subsystem. Five technologies are available. There are many details about the appropriate selection of the technology according to the IEC 61508 and IEC 61511 norms; the technologies T1 to T5 are hypotheticals and compatibles for heterogeneous redundancies, these technologies represent only different options. The subsystems analyzed have a 1-out-of- n structure, but with the appropriated equations others structures can be considered. Four redundancies are taking into account with homogeneous and heterogeneous (to avoid common cause failure) technology. The life cycle cost values were based on the cost information given in Torres-Echeverría et al. (2009) and Ramírez-Marengo et al. (2013). The present value factor (PVF) is evaluated using a discount rate (R) of 5% and the useful life of the system (n) of 10 years.

This case study is analyzed using six subcases. In the first subcase A, the problem is solved using only safety distance to risk reduction, without SIS or any active and passive measures. In the other subcases, different tolerable risk frequency, F_{TR}, values were used in order to

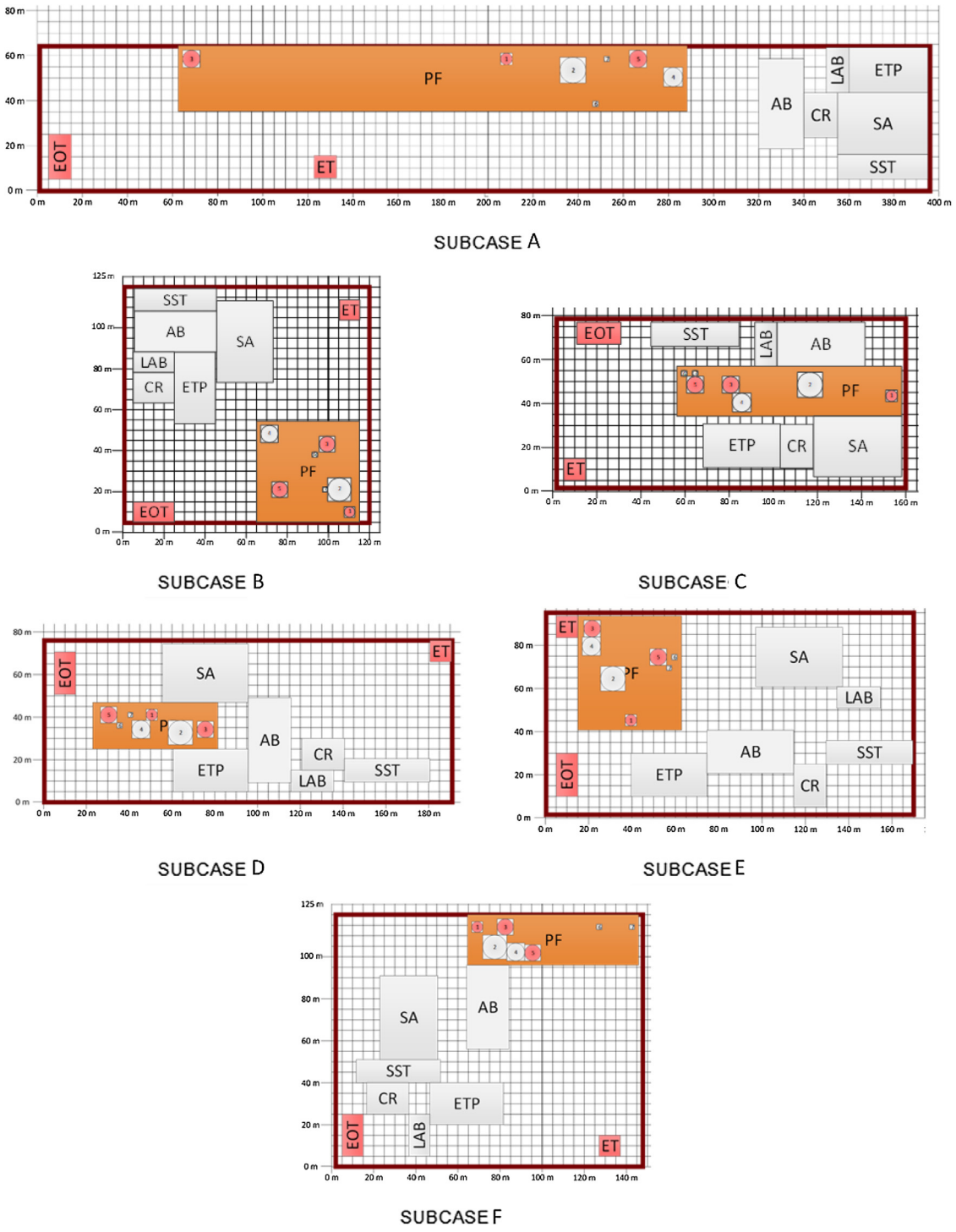


Fig. 6. Optimal Facility Layout results for subcases A, B, C, D, E and F.

test the mathematical model and provide different optimal layout. The F_{TR} values are 0.1, 0.01, 0.001, 0.0001 and 0.00,001 for the subcases B, C, D, E and F, respectively. Therefore, the F_{TR} value is a primary criterion to determine the optimal layout.

It is worth to mention that the model is a mixed-integer nonlinear programming (MINLP) problem, which consists of 5,292 continuous variables, 3,155 binary variables and 4,520 constraints; this model was implemented in the software GAMS 22.0 and solved through DICOPT (CONOPT/CPLEX) in a computer with an Intel Pentium CPU G645 processor at 2.90 GHz with 8 GB of RAM.

5. Results and discussions

This section contains an analysis of the results provided using the MINLP model. The results presented, in the following paragraphs, were obtained only considering the implementation of SIS, pipe and land costs, and the explosion effects; for more realistic results, additional information could be included in the mathematical model, for example, constraints about the particular requirements of the installation

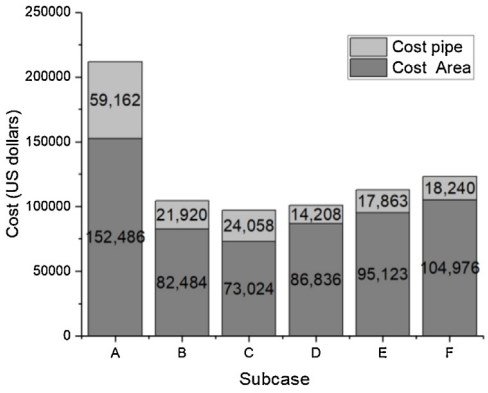


Fig. 7. Optimal layout for subcase A.

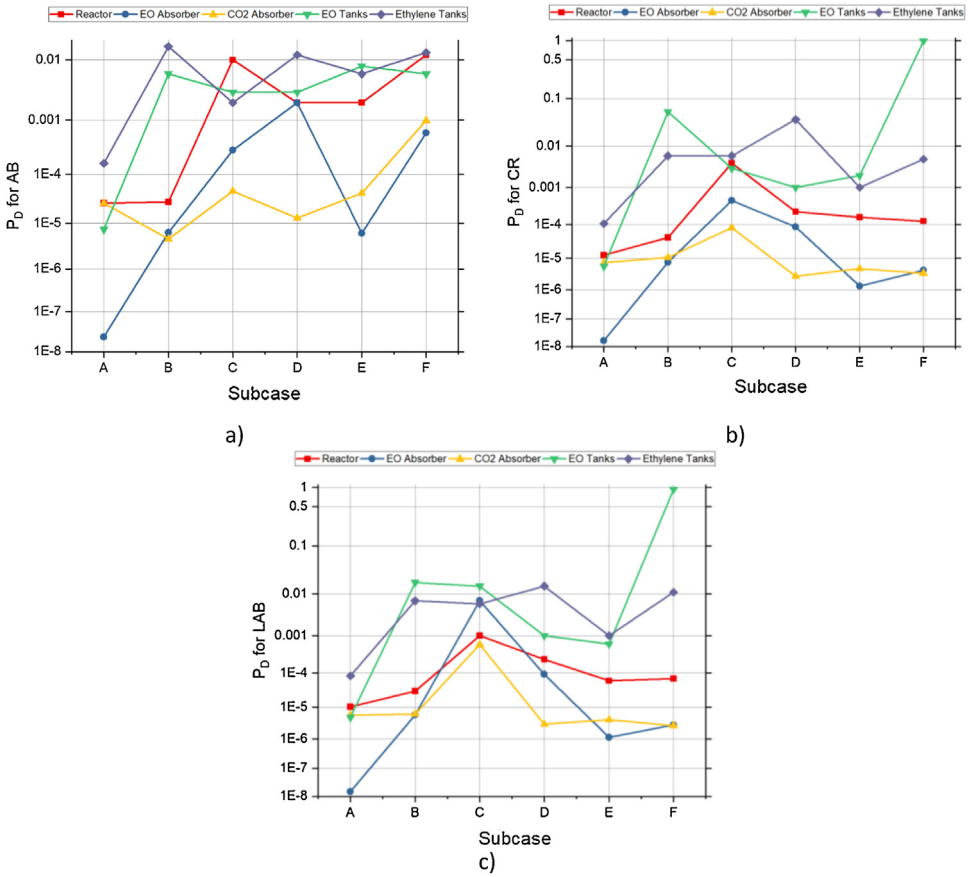


Fig. 8. Damage probability for a) administration building b) control room and c) laboratory.

of specific facilities. Also, the quality of the data provided to the model determines the quality of the solution. However, the appropriated application of this model could provide valuable additional information to the decision makers, avoiding the subjective point of view.

The Fig. 5 presents the total cost for each subcase. The absence of any device to protect the facilities produces high risk (yellow line) that represent the catastrophic damage; the minimum cost value under this restriction was for subcase A, because the optimization process achieved its value without an independent layer of protection, which provide a risk reduction factor. The domino effect produces these consequences. Then, the escalation events are essential to be avoided by reduction of their frequency, which is the purpose of the SIS installation for each hazardous equipment. For this reason, the domino effect risk determines the layout result. For subcases B to F, the yellow line points show the produced damage in the plant when the protection devices fail or are not installed. In these results, all the dangerous equipment is close to the other facilities. The total cost of the layout without protections is primarily due to the damages, but this cost is due to the SIS installation and operation costs when there are investments in process safety (red line). There are other benefits to invest in safety, like the image of the company. Dr. Trevor Kletz says, "There's an old saying that if you think safety is expensive, try an accident. Accidents cost a lot of money. And, not only in damage to plant and in claims for injury, but also in the loss of the company's reputation" (CSB-News, 2013). The damage cost of subcase A is 3.57 times the cost of subcase C, which is the best optimal solution founded. The total cost was \$8.22, \$2.31, \$2.30, \$2.38, \$3.84 and \$4.68 million US dollars for the subcases A, B, C, D, E, and F, respectively. These results were achieved for a specific F_{TR} value (0.1, 0.01, 0.001, 0.0001 and 0.00,001 for the subcases B, C, D, E, and F, respectively) that is a

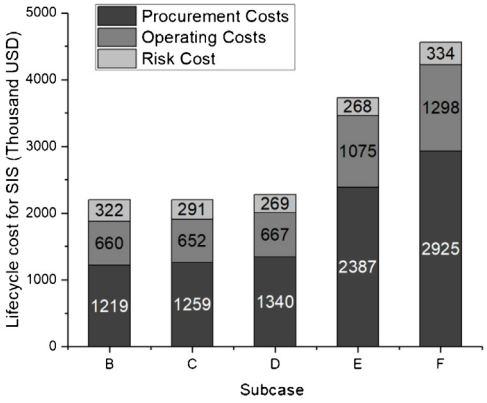


Fig. 9. Comparison of SIS lifecycle cost for each subcase.

Table 2
Failure rate per hour and beta factor data per subsystem and technology for reactor SIS.

Equipment:	Reactor											
Subsystem	Technology	$\frac{\lambda^{SD}}{1\times10^{-6}}$	$\frac{\lambda^{SU}}{1\times10^{-6}}$	$\frac{\lambda^{DD}}{1\times10^{-6}}$	$\frac{\lambda^{DU}}{1\times10^{-6}}$	$\frac{\lambda^{TO}}{1\times10^{-6}}$	SFF	β^{SU}	β^{SD}	β^{DU}	β^{DD}	M
Pressure sensor	T1	8.140	5.200	7.760	5.100	26.200	0.805	0.10	0.05	0.10	0.10	1.00
	T2	7.680	10.100	6.840	3.400	28.020	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.340	4.320	6.520	3.210	16.390	0.804	0.10	0.10	0.10	0.10	1.00
	T4	2.100	2.740	0.750	2.430	8.020	0.697	0.10	0.05	0.10	0.10	1.00
	T5	2.760	1.220	3.400	1.890	9.270	0.796	0.10	0.10	0.10	0.10	1.00
Temperature sensor	T1	9.840	4.660	5.320	8.330	28.150	0.704	0.10	0.02	0.10	0.10	1.00
	T2	3.650	3.340	3.460	5.230	15.680	0.666	0.10	0.05	0.10	0.10	1.00
	T3	7.430	5.560	1.540	3.440	17.970	0.809	0.10	0.10	0.10	0.10	1.00
	T4	1.130	1.860	1.230	2.350	6.570	0.642	0.10	0.02	0.10	0.10	1.00
	T5	0.500	1.360	3.220	1.230	6.310	0.805	0.10	0.05	0.10	0.10	1.00
Logic Solver	T1	5.640	3.860	2.780	5.500	17.780	0.691	0.10	0.10	0.10	0.10	1.00
	T2	8.100	3.340	0.000	2.560	14.000	0.817	0.10	0.10	0.10	0.10	1.00
	T3	0.840	1.350	0.400	1.350	3.940	0.657	0.10	0.10	0.10	0.10	1.00
	T4	1.306	1.188	1.234	1.081	4.808	0.775	0.10	0.10	0.10	0.10	1.00
	T5	1.223	1.360	1.501	0.850	4.934	0.828	0.10	0.10	0.10	0.10	1.00
Actuator	T1	4.200	4.300	1.340	4.230	14.070	0.699	0.10	0.10	0.10	0.10	1.00
	T2	1.234	4.010	0.540	3.540	9.324	0.620	0.10	0.10	0.10	0.10	1.00
	T3	2.440	2.430	1.200	2.650	8.720	0.696	0.10	0.10	0.10	0.10	1.00
	T4	1.650	1.660	1.600	2.221	7.131	0.689	0.10	0.10	0.10	0.10	1.00
	T5	1.544	0.700	1.760	0.860	4.864	0.823	0.10	0.10	0.10	0.10	1.00

constraint to satisfy during the optimization process. However, when the minimization of the total cost was solved without this constraint, the layout solution was the same result of subcase C. Therefore, when the optimization problem is not subject to any F_{TR} value, the optimal solution will be better or equal to the best solution obtained by adding this F_{TR} constraint (Eq. (73)), but the F_{TR} value is the responsibility of the decision-makers.

The facility layout representation for each subcase is showed in the Fig. 6 and Table 12 contains the coordinates-(x,y) value of the equipment and facilities. These solutions are local optimum, the non-convexities of the model difficult the possibility to find or guarantee the global solution. Therefore, the global optimization of the layout problem is still a challenge for academics. The total cost for subcases B and C are similar, according to the Fig. 5. Two differences are the land and pipeline costs, Fig. 7. The pipe cost for subcase C is lower than subcase B, but the land cost for subcase C is more significant than subcase B. However, the result of combine both costs is better for subcase C. The pipeline cost for subcase C is high because the distance between the reactor and the heat exchanger A and the CO_2 absorber is more significant in comparison to the process layout of subcase B, but the lowest pipe cost was for subcase D. Also, the dimensions for the processing facility and the plant are reported in Table 13. The occupied area for the process is not reduced when the probability of the F_{TR} value is reduced. The lowest value of the process area was for subcase D (a reason why the pipe cost is also the minimum value of these six subcases) and for the minimum plant area value was for subcase C. Therefore, the layouts for subcases B and C are the most favorable for the economic aspect. The next paragraphs contain an evaluation from the safety point of view.

The locations of the administration buildings (AB), control room (CR) and laboratory (LAB) are crucial to guarantee the safety of the personal. Fig. 8 shows the damage probability for these facilities. In the subcase A, the ethylene vessel produces the maximum damage probability over the administration building, control room, and laboratory with the values 1.62×10^{-4}, 1.06×10^{-4} and 8.36×10^{-5}, respectively. These buildings are located in the opposite direction of the dangerous equipment. The others five subcases considering the protection by SIS, but these damage probabilities were evaluated without the SIS reduction frequency because the objective was determinate the damage if the explosion occurs by SIS failure. For subcase B, the ethylene vessels produce the main damage probability for the administration building, 0.016. The control room, 0.055, and the laboratory, 0.018, are primarily affected by the explosion of ethylene oxide vessel. For subcase C, the reactor produces the main affectation over the administration buildings with a damage probability of 0.01, the ethylene vessel for the control room 0.003, and ethylene oxide for the laboratory 0.015. For the subcases, B to E, the damage probability for the administration buildings takes values in the interval 0.8% and 1.6%, but the damage probabilities for subcase F are 99% and 93% for

Table 3
Failure rate per hour and beta factor data per subsystem and technology for ethylene oxide absorber SIS.

Equipment:	Ethylene oxide absorber											
Subsystem	Technology	$\frac{\lambda^{SD}}{1\times10^{-6}}$	$\frac{\lambda^{SU}}{1\times10^{-6}}$	$\frac{\lambda^{DD}}{1\times10^{-6}}$	$\frac{\lambda^{DU}}{1\times10^{-6}}$	$\frac{\lambda^{TO}}{1\times10^{-6}}$	*SFF*	β^{SU}	β^{SD}	β^{DU}	β^{DD}	*M*
Pressure sensor	T1	7.012	4.420	7.213	4.335	22.980	0.805	0.10	0.05	0.10	0.10	1.00
	T2	6.850	8.585	5.778	2.890	24.103	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.005	3.672	5.453	2.729	13.859	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.825	2.329	0.348	2.066	6.567	0.697	0.10	0.05	0.10	0.10	1.00
	T5	2.236	1.037	3.003	1.607	7.882	0.796	0.10	0.10	0.10	0.10	1.00
Temperature sensor	T1	8.365	3.961	5.010	7.081	24.416	0.805	0.10	0.02	0.10	0.10	1.00
	T2	3.050	2.839	3.121	4.446	13.456	0.879	0.10	0.05	0.10	0.10	1.00
	T3	6.310	4.726	1.556	2.924	15.516	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.002	1.581	1.215	1.998	5.796	0.697	0.10	0.02	0.10	0.10	1.00
	T5	0.430	1.156	2.546	1.046	5.178	0.796	0.10	0.05	0.10	0.10	1.00
Logic Solver	T1	5.121	3.281	2.646	4.675	15.723	0.805	0.10	0.10	0.10	0.10	1.00
	T2	7.012	2.839	0.223	2.176	12.250	0.879	0.10	0.10	0.10	0.10	1.00
	T3	0.690	1.148	0.340	1.148	3.325	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.100	1.009	1.066	0.919	4.094	0.697	0.10	0.10	0.10	0.10	1.00
	T5	1.002	1.156	0.999	0.723	3.879	0.796	0.10	0.10	0.10	0.10	1.00
Actuator	T1	3.800	3.655	1.200	3.596	12.251	0.805	0.10	0.10	0.10	0.10	1.00
	T2	1.055	3.409	0.464	3.009	7.936	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.084	2.066	1.056	2.253	7.458	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.403	1.411	0.146	1.888	4.847	0.697	0.10	0.10	0.10	0.10	1.00
	T5	1.313	0.595	1.510	0.731	4.149	0.796	0.10	0.10	0.10	0.10	1.00

Table 4
Failure rate per hour and beta factor data per subsystem and technology for carbon dioxide absorber SIS.

Equipment:	Carbon dioxide absorber											
Subsystem	Technology	$\frac{\lambda^{SD}}{1\times10^{-6}}$	$\frac{\lambda^{SU}}{1\times10^{-6}}$	$\frac{\lambda^{DD}}{1\times10^{-6}}$	$\frac{\lambda^{DU}}{1\times10^{-6}}$	$\frac{\lambda^{TO}}{1\times10^{-6}}$	*SFF*	β^{SU}	β^{SD}	β^{DU}	β^{DD}	*M*
Pressure sensor	T1	7.420	7.076	6.898	3.931	25.326	0.811	0.10	0.05	0.10	0.10	1.00
	T2	4.719	6.669	6.148	3.057	20.593	0.880	0.10	0.10	0.10	0.10	1.00
	T3	4.109	4.711	4.547	2.899	16.266	0.803	0.10	0.10	0.10	0.10	1.00
	T4	2.068	3.265	3.268	2.609	11.209	0.685	0.10	0.05	0.10	0.10	1.00
	T5	2.230	1.832	1.875	1.998	7.935	0.796	0.10	0.05	0.10	0.10	1.00
Temperature sensor	T1	6.226	3.700	4.228	6.272	20.426	0.710	0.10	0.02	0.10	0.10	1.00
	T2	5.110	4.116	2.419	4.010	15.655	0.670	0.10	0.05	0.10	0.10	1.00
	T3	4.017	3.184	2.521	3.654	13.377	0.812	0.10	0.10	0.10	0.10	1.00
	T4	3.968	3.432	1.385	2.678	11.463	0.655	0.10	0.02	0.10	0.10	1.00
	T5	0.765	1.489	2.053	1.656	5.963	0.798	0.10	0.02	0.10	0.10	1.00
Logic Solver	T1	6.468	3.330	1.412	3.728	14.938	0.703	0.10	0.10	0.10	0.10	1.00
	T2	4.160	2.169	0.241	1.808	8.379	0.822	0.10	0.10	0.10	0.10	1.00
	T3	3.193	1.963	0.903	1.888	7.947	0.655	0.10	0.10	0.10	0.10	1.00
	T4	0.984	1.174	0.760	1.124	4.042	0.776	0.10	0.10	0.10	0.10	1.00
	T5	1.158	1.178	1.200	0.893	4.429	0.814	0.10	0.02	0.10	0.10	1.00
Actuator	T1	2.572	3.843	0.886	3.594	10.895	0.707	0.10	0.10	0.10	0.10	1.00
	T2	1.703	2.979	0.815	2.863	8.360	0.621	0.10	0.10	0.10	0.10	1.00
	T3	1.912	2.451	0.989	2.534	7.885	0.698	0.10	0.10	0.10	0.10	1.00
	T4	1.894	1.892	1.001	2.253	7.039	0.611	0.10	0.10	0.10	0.10	1.00
	T5	1.477	1.092	1.254	1.425	5.248	0.824	0.10	0.10	0.10	0.10	1.00

the control room and laboratory, respectively. For subcases D and E, the probability damage of the control room does not exceed 0.038 and 0.015 for the laboratory. In consequence, the layout result for subcase F has more disadvantage because it is the most expensive; it is worth to remember that is not a global optimum solution and different layout under the actual constraints could be found with different initial values or more computational time. In subcase F, the Administration Buildings are located in the middle of the land trying to reduce the effects of the explosion produced by dangerous equipment located around the place.

The results show that the increase in the cost of the SIS does not reduce the cost of the land and pipe occupied; except for subcase C, the control room is remote from the process area in order to protect the workers inside of this facility. Nevertheless, if this location has operational problems, then additional independent layers of protection can be added to reduce the risk over this facility. Thus, the risk reduction factor of these non-SIS layers of protection is included in the $PFD_i^{NON-SIS}$ term to avoid these separations.

The SIS cost is presented in Fig. 9; these results depend on the cost data provided to the mathematical model, Tables 2–11. In this case, the main cost of the SIS is the initial inversion. The procurement cost represents from 55% to 64% of the SIS cost, the operational cost values are between the 28% to 30% of the cost, and the risk cost is between 7% to 15% of the total cost. There is a relationship between the procurement cost with risk cost, when the risk cost is high the initial costs are low and the other way around, but the operational cost does not change substantially. The minimum SIS cost was for subcase B. The subcases C, D, E, F are 0.11%, 3.4%, 69.5% and 107% more expensive that subcase B, respectively. Therefore, subcases B and C have, practically, the same SIS cost.

Fig. 10 shows a comparison of the operating and risk costs. The total operational cost combines preventive and corrective maintenance and testing costs. Preventive maintenance is the most relevant cost, between the 84–87 percent of the total operating cost, and it increases when the F_{TR} is reduced. The lowest percentage is produced for subcase B, and the highest percentage for subcase C. The corrective

Table 5
: Failure rate per hour and beta factor data per subsystem and technology for ethylene oxide vessel SIS.

Equipment:	Ethylene oxide vessel											
Subsystem	Technology	$\frac{\lambda^{SD}}{1\times10^{-6}}$	$\frac{\lambda^{SU}}{1\times10^{-6}}$	$\frac{\lambda^{DD}}{1\times10^{-6}}$	$\frac{\lambda^{DU}}{1\times10^{-6}}$	$\frac{\lambda^{TO}}{1\times10^{-6}}$	*SFF*	β^{SU}	β^{SD}	β^{DU}	β^{DD}	*M*
Pressure sensor	T1	7.609	5.341	7.209	4.637	24.796	0.805	0.10	0.04	0.10	0.10	1.00
	T2	6.725	8.959	6.400	3.178	25.263	0.879	0.10	0.06	0.10	0.10	1.00
	T3	2.177	3.915	5.595	2.945	14.632	0.804	0.10	0.30	0.10	0.10	1.00
	T4	2.035	2.435	0.950	2.234	7.654	0.697	0.10	0.05	0.10	0.10	1.00
	T5	2.084	0.921	2.567	1.427	6.999	0.796	0.10	0.10	0.10	0.10	1.00
Temperature sensor	T1	8.677	4.258	4.829	7.543	25.308	0.805	0.10	0.02	0.10	0.10	1.00
	T2	3.781	3.345	3.075	4.751	14.951	0.879	0.10	0.10	0.10	0.10	1.00
	T3	6.401	4.884	1.419	3.133	15.837	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.004	1.702	1.340	2.105	6.152	0.697	0.10	0.30	0.10	0.10	1.00
	T5	0.378	1.027	2.431	0.929	4.764	0.796	0.10	0.40	0.10	0.10	1.00
Logic Solver	T1	5.529	3.580	2.356	4.898	16.364	0.805	0.10	0.10	0.10	0.10	1.00
	T2	6.942	2.956	0.037	2.294	12.229	0.879	0.10	0.10	0.10	0.10	1.00
	T3	0.833	1.254	0.453	1.244	3.784	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.220	1.132	1.184	0.995	4.531	0.697	0.10	0.10	0.10	0.10	1.00
	T5	0.924	1.027	1.133	0.642	3.725	0.796	0.10	0.10	0.10	0.10	1.00
Actuator	T1	3.674	4.015	1.186	3.912	12.787	0.805	0.10	0.10	0.10	0.10	1.00
	T2	1.272	3.623	0.569	3.245	8.709	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.221	2.213	1.165	2.451	8.050	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.541	1.472	1.519	1.962	6.493	0.697	0.10	0.10	0.10	0.10	1.00
	T5	1.166	0.529	1.329	0.649	3.673	0.796	0.10	0.10	0.10	0.10	1.00

Table 6
: Failure rate per hour and beta factor data per subsystem and technology for ethylene vessel SIS.

Equipment:	Ethylene vessel											
Subsystem	Technology	$\frac{\lambda^{SD}}{1\times10^{-6}}$	$\frac{\lambda^{SU}}{1\times10^{-6}}$	$\frac{\lambda^{DD}}{1\times10^{-6}}$	$\frac{\lambda^{DU}}{1\times10^{-6}}$	$\frac{\lambda^{TO}}{1\times10^{-6}}$	*SFF*	β^{SU}	β^{SD}	β^{DU}	β^{DD}	*M*
Pressure sensor	T1	7.543	5.635	7.115	4.504	24.796	0.805	0.10	0.05	0.10	0.10	1.00
	T2	6.345	8.540	6.349	3.151	24.384	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.151	3.793	5.192	2.880	14.017	0.804	0.10	0.10	0.10	0.10	1.00
	T4	2.068	2.324	1.118	2.188	7.699	0.697	0.10	0.10	0.10	0.10	1.00
	T5	1.714	0.758	2.111	1.174	5.757	0.796	0.10	0.10	0.10	0.10	1.00
Temperature sensor	T1	8.232	4.152	4.684	7.305	24.373	0.805	0.10	0.10	0.10	0.10	1.00
	T2	4.009	3.474	2.935	4.611	15.029	0.879	0.10	0.10	0.10	0.10	1.00
	T3	5.960	4.620	1.392	3.047	15.018	0.804	0.10	0.10	0.10	0.10	1.00
	T4	0.958	1.661	1.464	2.022	6.105	0.697	0.10	0.10	0.10	0.10	1.00
	T5	0.311	0.845	2.000	0.764	3.919	0.796	0.10	0.10	0.10	0.10	1.00
Logic Solver	T1	5.663	3.529	2.164	4.683	16.039	0.805	0.10	0.10	0.10	0.10	1.00
	T2	6.436	2.812	0.063	2.205	11.515	0.879	0.10	0.10	0.10	0.10	1.00
	T3	0.859	1.237	0.505	1.221	3.822	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.208	1.138	1.196	0.975	4.517	0.697	0.10	0.10	0.10	0.10	1.00
	T5	0.760	0.845	0.932	0.528	3.064	0.796	0.10	0.10	0.10	0.10	1.00
Actuator	T1	3.463	3.977	1.128	3.849	12.416	0.805	0.10	0.10	0.10	0.10	1.00
	T2	1.344	3.503	0.609	3.172	8.628	0.879	0.10	0.10	0.10	0.10	1.00
	T3	2.158	2.152	1.186	2.412	7.908	0.804	0.10	0.10	0.10	0.10	1.00
	T4	1.527	1.402	1.522	1.864	6.315	0.697	0.10	0.10	0.10	0.10	1.00
	T5	0.959	0.435	1.093	0.534	3.021	0.796	0.10	0.10	0.10	0.10	1.00

Table 7
Life cycle cost data for reactor SIS.

Subsystem	Technology ($)	Purchase Cost ($)	Design Cost ($)	Installation / Consumption Cost ($)	Preventive Maintenance Cost ($)	Repair Cost Per Instrument ($-Event)	Test Interval Maintenance Cost ($)
Pressure sensor	T1	35,000	900	430	4,200	4,200	100
	T2	50,000	900	440	4,400	4,400	75
	T3	53,000	900	450	4,320	4,320	85
	T4	58,000	900	450	4,500	4,500	80
	T5	65,000	900	460	4,350	4,350	80
Temperature sensor	T1	32,000	900	430	4,400	4,400	120
	T2	40,000	900	440	4,400	4,400	100
	T3	50,000	900	450	4,450	4,450	89
	T4	59,000	900	450	4,600	4,600	65
	T5	68,000	900	460	4,300	4,300	90
Logic Solver	T1	100,000	1,200	770	3,500	10,050	250
	T2	120,000	1,200	770	3,700	12,050	130
	T3	130,000	1,200	770	3,500	13,050	140
	T4	138,000	1,200	800	3,750	13,850	100
	T5	145,000	1,200	800	3,500	14,550	100
Actuator	T1	35,000	900	430	4,000	3,550	120
	T2	38,000	900	440	4,300	3,850	100
	T3	41,500	900	450	4,200	4,200	89
	T4	45,000	900	450	3,900	4,550	65
	T5	55,000	900	460	3,900	5,550	90

Table 8
Life cycle cost data for ethylene oxide absorber SIS.

Subsystem	Technology ($)	Purchase Cost ($)	Design Cost ($)	Installation / Consumption Cost ($)	Preventive Maintenance Cost ($)	Repair Cost Per Instrument ($-Event)	Test Interval Maintenance Cost ($)
Pressure sensor	T1	30,000	800	430	3,500	3,570	85
	T2	42,500	800	440	3,700	3,740	65
	T3	45,050	800	450	3„700	3,670	75
	T4	49,300	800	450	3800	3,800	70
	T5	50,000	800	460	3,700	3,700	70
Temperature sensor	T1	28,000	800	430	3,740	3,740	100
	T2	34,000	800	440	3,740	3,740	85
	T3	43,000	800	450	3,800	3,800	76
	T4	50,000	800	450	3,900	3,900	55
	T5	58,000	800	460	3,700	3,655	77
Logic Solver	T1	90,000	1,020	770	3,000	8,500	212
	T2	110,000	1,020	770	3,200	10,200	110
	T3	120,000	1,020	770	3,000	12,000	120
	T4	125,000	1,020	800	3,200	12,500	85
	T5	130,000	1,020	800	3,000	13,000	85
Actuator	T1	30,000	800	430	3,400	3,000	102
	T2	33,000	800	440	3,700	3,200	85
	T3	35,000	800	450	3,600	3,600	76
	T4	38,000	800	450	3,300	3,900	55
	T5	47,000	800	460	3,300	4,700	77

Table 9
Life cycle cost data for carbon dioxide absorber SIS.

Subsystem	Technology ($)	Purchase Cost ($)	Design Cost ($)	Installation / Consumption Cost ($)	Preventive Maintenance Cost ($)	Repair Cost Per Instrument ($-Event)	Test Interval Maintenance Cost ($)
Pressure sensor	T1	27,000	800	430	3,500	3,200	73
	T2	38,250	800	440	3,700	3,400	65
	T3	37,500	800	445	3,675	3,330	71
	T4	40,545	800	450	3,700	3,300	75
	T5	44,370	800	450	3,800	3,420	70
Temperature sensor	T1	25,200	800	430	3,179	3,360	100
	T2	30,600	800	440	3,179	3,360	73
	T3	34,875	800	442	3,520	3,412	76
	T4	38,700	800	450	3,800	3,420	76
	T5	45,000	800	450	3,900	3,510	55
Logic Solver	T1	81,000	867	770	3,000	7,650	212
	T2	99,000	867	770	3,200	9,180	110
	T3	10,0125	867	780	3,100	9,720	130
	T4	108,000	867	770	3,000	10,800	120
	T5	112,500	867	800	3,200	11,250	73
Actuator	T1	27,000	800	430	2,890	2,700	87
	T2	29,700	800	440	3,700	2,880	72
	T3	30,675	800	442	3,372	3,082	72
	T4	32,000	800	450	3,600	3,240	76
	T5	34,000	800	450	3,300	3,510	55

Table 10
Life cycle cost data for ethylene oxide vessel SIS.

Subsystem	Technology ($)	Purchase Cost ($)	Design Cost ($)	Installation / Consumption Cost ($)	Preventive Maintenance Cost ($)	Repair Cost Per Instrument($-Event)	Test Interval Maintenance Cost ($)
Pressure sensor	T1	38,500	900	430	3,780	5,040	80
	T2	55,000	900	440	3,960	5,280	60
	T3	58,300	900	450	3,888	5,184	68
	T4	63,800	900	450	4,050	5,400	64
	T5	71,500	900	460	3,915	5,220	64
Temperature sensor	T1	35,200	900	430	3,960	5,280	96
	T2	44,000	900	440	3,960	5,280	80
	T3	55,000	900	450	4,005	5,340	70
	T4	64,900	900	450	4,140	5,520	52
	T5	74,800	900	460	3,870	5,160	72
Logic Solver	T1	110,000	1,200	770	3,150	12,060	200
	T2	132,000	1,200	770	3,330	14,460	104
	T3	143,000	1,200	770	3,150	15,660	112
	T4	151,800	1,200	800	3,375	16,620	80
	T5	159,500	1,200	800	3,150	17,460	80
Actuator	T1	38,500	900	430	3,600	4,260	96
	T2	41,800	900	440	3,870	4,620	80
	T3	45,650	900	450	3,780	5,040	70
	T4	49,500	900	450	3,510	5,460	52
	T5	60,500	900	460	3,510	6,660	72

Table 11
Life cycle cost data for ethylene vessel SIS.

Subsystem	Technology ($)	Purchase Cost ($)	Design Cost ($)	Installation / Consumption Cost ($)	Preventive Maintenance Cost ($)	Repair Cost Per Instrument($-Event)	Test Interval Maintenance Cost ($)
Pressure sensor	T1	36,575	900	430	3,800	4,536	96
	T2	52,250	900	440	4,020	4,752	72
	T3	55,385	900	450	4,000	4,666	82
	T4	60,610	900	450	4,100	4,860	77
	T5	67,925	900	460	4,000	4,698	77
Temperature sensor	T1	33,440	900	430	4,050	4,752	115
	T2	41,800	900	440	4,050	4,752	96
	T3	52,250	900	450	4,085	4,806	84
	T4	61,655	900	450	4,200	4,968	62
	T5	71,060	900	460	4,000	4,644	86
Logic Solver	T1	104,500	1,200	770	3,200	10,854	240
	T2	125,400	1,200	770	3,410	13,014	125
	T3	135,850	1,200	770	3,200	14,094	134
	T4	144,210	1,200	800	4,000	14,958	96
	T5	151,525	1,200	800	3,200	15,714	96
Actuator	T1	36,575	900	430	3,700	3,834	115
	T2	39,710	900	440	4,000	4,158	96
	T3	43,000	900	450	3,860	4,536	84
	T4	47,025	900	450	3,570	4,914	62
	T5	57,475	900	460	3,570	5,994	86

Table 12
Optimal solutions - locations.

Unit	A		B		C		D		E		F	
	x	*y*	*x*	*y*	*x*	*y*	*x*	*y*	*x*	*y*	*x*	*y*
Process-Facility	175.4	49.8	90.1	29.7	107.2	45.6	48.2	35.9	39.0	67.1	105.2	108.0
Control-Room	347.5	33.5	15.0	70.7	110.7	20.6	130.6	22.5	122.1	15.0	26.7	32.5
Effluent-Treatment-Plant	377.5	53.5	35.0	70.7	85.7	20.8	78.1	15.0	57.1	20.0	64.2	30.0
Ethylene Oxide Tanks	10.0	15.0	15.0	10.0	21.0	71.9	10.0	60.6	10.0	20.0	10.0	15.0
Ethylene Tanks	127.7	10.5	110.3	108.7	10.0	10.0	185.6	70.9	10.0	88.5	132.1	10.0
Solvent Storage Tanks	375.0	10.5	25.5	113.7	64.6	71.4	160.6	15.0	149.6	30.4	31.7	45.5
Storage Area	375.0	29.8	59.4	93.2	138.2	20.4	75.6	60.6	117.1	74.6	36.7	71.0
Administration Buildings	330.0	38.5	25.6	98.2	121.6	66.9	105.6	29.2	94.6	30.7	74.2	76.0
Laboratory	355.0	53.5	15.0	83.2	96.6	66.9	125.6	10.0	144.6	55.8	41.7	15.0
Reactor	208.1	58.5	110.3	10.0	153.6	43.5	50.6	41.0	39.5	45.3	69.2	114.2
Heat Exchanger A	237.6	53.5	105.3	21.0	116.6	48.5	63.9	32.7	31.2	64.5	77.5	104.6
Ethylene Oxide Absorber	68.3	58.5	99.4	43.2	80.7	48.5	75.6	34.2	21.6	87.6	82.5	114.2
Heat Exchanger B	282.1	50.4	71.3	48.2	85.7	40.4	45.6	34.2	21.2	79.6	87.4	102.2
CO_2 Absorber	266.5	58.5	76.3	21.0	64.6	48.5	30.4	41.0	52.1	74.6	95.5	101.8
Flash Tank	247.6	38.5	93.4	38.0	59.6	53.6	35.6	36.0	59.6	74.5	127.1	114.2
Pump	252.6	58.5	98.4	21.0	64.6	53.7	40.6	41.0	57.1	69.5	142.6	114.2

Table 13
Optimal solutions – Land occupied, length and width of the process and plant area.

Variable	Subcase					
	A	B	C	D	E	F
$l^x_{Process}$ (m)	225.8	50.0	101.9	66.5	47.9	81.2
$l^y_{Process}$ (m)	29.1	49.4	22.8	21.9	52.8	24.0
Process area (m^2)	6,581	2,473	2,320	1,454	2,530	1,950
Dplant (*m*)	64.3	119.2	76.9	75.9	93.5	120.0
Lplant (*m*)	395.0	115.3	158.2	190.6	169.6	145.8
AreaPlant (m^2)	25,414.3	13,747.3	12,170.6	14,472.7	15,853.9	17,495.9

maintenance represents between the 11–14 percent (lowest for subcase E and the highest for subcase B), and the testing cost is only the 2 or 3 percent of the operational cost.

On the other hand, the safety instrumented systems installed for subcase D produce the minimum spurious trip cost and the subcase F the minimum hazard cost. When the F_{TR} is reduced the spurious trip cost and the hazard cost are reduced for subcases B, C, and D, but for subcases E and F both costs are in the opposite direction, i.e., the spurious trips increase, and the hazard cost is reduced.

The PFD_i, the SIL and the *STR* per SIS is presented in Table 14. The F_{TR} for subcases B and C were 0.1 and 0.01. However, SIL 2 is required for ethylene oxide and ethylene vessels in order to reduce the risk cost, that allows reducing the pipeline and land cost too. For subcases B and C, the reactor and both absorbers have a SIL 1. For these reasons, all the costs for both subcases are very similar, also the land and the pipe costs. The safety instrumented systems for the reactor and the ethylene oxide absorber have lower probability of failure on demand value for subcase C than subcase B. Table 15 shows the probability of failure on demand per subsystem; this table includes information about the architecture and the technology selected in each subcase. This data provides more details about the selected elements. For example, the safety instrumented systems for the ethylene vessel is the same for subcases B and C; also, this happened in the ethylene oxide vessel too.

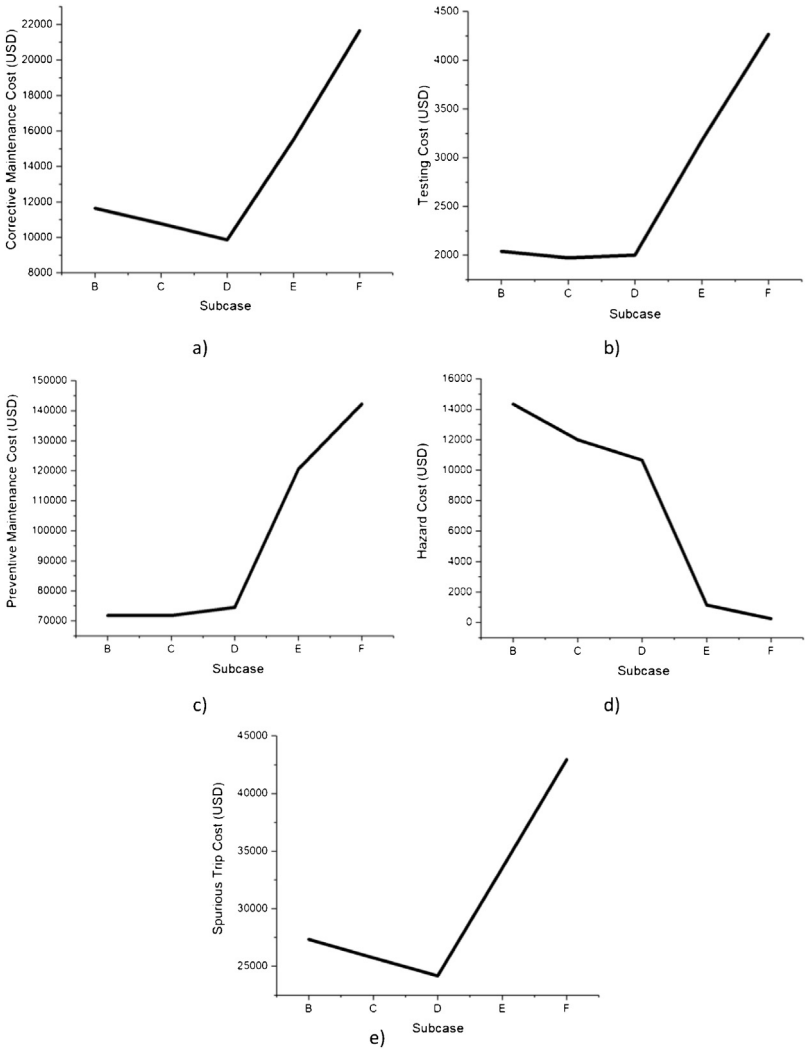


Fig. 10. Comparison of a) corrective maintenance, b) testing, c) preventive maintenance, d) hazard and e) spurious trip costs for the optimal solution of subcases B to F.

Table 14
Optimal solutions - Probability of Failure on Demand for SIS, Safety Integrity Levels, and Spurious Trip Rate.

Unit	B			C			D			E			F		
	PFD_i^{SIS}	SIL	STR_i	PFD_i^{SIS}	SIL	STR_i	PFD_i^{SIS}	SIL	STR_i	PFD_i^{SIS}	SIL	STR_i	PFD_i^{SIS}	SIL	STR_i
Reactor	1.6×10^{-2}	1	0.226	1.0×10^{-2}	1	0.217	1.0×10^{-2}	1	0.142	9.6×10^{-4}	3	0.314	8.0×10^{-5}	4	0.348
EO Absorber	2.9×10^{-2}	1	0.250	1.4×10^{-2}	1	0.192	9.0×10^{-3}	2	0.185	9.1×10^{-4}	3	0.307	6.4×10^{-5}	4	0.289
CO_2 Absorber	2.3×10^{-2}	1	0.242	2.3×10^{-2}	1	0.242	7.0×10^{-3}	2	0.259	8.4×10^{-4}	3	0.287	8.5×10^{-5}	4	0.333
EO vessels	9.0×10^{-3}	2	0.209	9.0×10^{-3}	2	0.209	9.0×10^{-3}	2	0.209	9.3×10^{-4}	3	0.345	8.6×10^{-5}	4	0.381
Ethylene vessels	8.0×10^{-3}	2	0.212	8.0×10^{-3}	2	0.212	8.0×10^{-3}	2	0.212	9.1×10^{-4}	3	0.145	7.0×10^{-5}	4	0.439

All the SIS components for subcase B and C have 1oo1 architecture; however, the actuator requires technology T4 and T5 (in this study T5 is more reliable than T4, T4 is more reliable than T3, and so on). For subcase D, the reactor has a SIL 1, but the others equipment has a SIL 2 to achieve F_{TR} below to 0.001. The equipment for subcase E need a SIL 3 and for subcase F a SIL 4. The selected architecture changes when more protection is required. The architecture 1oo1 is the most common. All the pressure and temperature sensors have architecture 1oo1 with two and one exceptions, respectively. For subcase E, the absorber and vessels for ethylene oxide have architecture 1oo2 for the pressure sensor, and the reactor has a temperature sensor with the same architecture. The controller has a 1-out-of-n structure, for subcases B, C, and D the structure is 1oo1 without homogenous redundancy, but in subcases E and F 1oo2 and 1oo3 homogenous redundancy is selected, respectively. Moreover, it is well known that the actuator is the critical element, because its mechanic components present the primary failure rate from the SIS; for this reason, the actuator from subcase A uses technology T4 and T5 and their complexity is increasing while the F_{TR} is reduced, even heterogeneous redundancy is selected for subcase F with structure 1oo3. The controller in subcase F also has a heterogeneous redundancy. Finally, the sensors do not present this structure because both sensors work together in detecting the irregular condition of the process (like redundancy), i.e., the controllers and valves require a better technology that the sensors for these cases.

In conclusion, the minimum cost of the layout, for the analyzed case study, is given by subcase C; this result was also achieved solving the problem without any tolerable risk frequency criteria. However, additional information can be obtained by the solution of other subcases

Table 15
Optimal solutions - Probability of Failure on Demand per Subsystem (A = Architecture; and T = Technology).

Subcase	Unit	Pressure Sensor			Temperature Sensor			Controller			Valve		
		PDF	A	T	PDF	A	T	PDF	A	T	PDF	A	T
	Reactor	2.2×10^{-2}	1oo1	T1	2.3×10^{-2}	1oo1	T2	6.0×10^{-3}	1oo1	T3	1.0×10^{-2}	1oo1	T4
	EO Absorber	1.9×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	2.1×10^{-2}	1oo1	T1	8.0×10^{-3}	1oo1	T4
B	CO_2 Absorber	1.1×10^{-2}	1oo1	T4	2.8×10^{-2}	1oo1	T1	1.6×10^{-2}	1oo1	T1	6.0×10^{-3}	1oo1	T5
	EO vessels	2.0×10^{-2}	1oo1	T1	2.1×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	3.0×10^{-3}	1oo1	T5
	Ethylene vessels	2.0×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	2.0×10^{-3}	1oo1	T5
	Reactor	2.2×10^{-2}	1oo1	T1	2.3×10^{-2}	1oo1	T2	6.0×10^{-3}	1oo1	T3	4.0×10^{-3}	1oo1	T5
	EO Absorber	1.9×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	8.0×10^{-3}	1oo1	T4
C	CO_2 Absorber	1.1×10^{-2}	1oo1	T1	2.8×10^{-2}	1oo1	T1	1.6×10^{-2}	1oo1	T4	6.0×10^{-3}	1oo1	T5
	EO vessels	2.0×10^{-2}	1oo1	T1	2.1×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	3.0×10^{-3}	1oo1	T5
	Ethylene vessels	2.0×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	2.0×10^{-3}	1oo1	T5
	Reactor	1.1×10^{-2}	1oo1	T4	2.3×10^{-2}	1oo1	T2	6.0×10^{-3}	1oo1	T3	4.0×10^{-3}	1oo1	T5
	EO Absorber	1.9×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	3.0×10^{-3}	1oo1	T5
D	CO_2 Absorber	1.1×10^{-2}	1oo1	T4	2.8×10^{-2}	1oo1	T1	5.0×10^{-3}	1oo1	T4	2.0×10^{-3}	1oo2	T1
	EO vessels	2.0×10^{-2}	1oo1	T1	2.1×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	3.0×10^{-3}	1oo1	T5
	Ethylene vessels	2.0×10^{-2}	1oo1	T1	2.0×10^{-2}	1oo1	T2	5.0×10^{-3}	1oo1	T3	2.0×10^{-3}	1oo1	T5
	Reactor	2.2×10^{-2}	1oo1	T1	3.0×10^{-3}	1oo2	T2	5.0×10^{-4}	1oo2	T4	3.9×10^{-4}	1oo2	T5
	EO Absorber	2.0×10^{-3}	1oo2	T1	2.0×10^{-2}	1oo1	T1	5.3×10^{-4}	1oo2	T3	3.3×10^{-4}	1oo2	T5
E	CO_2 Absorber	1.1×10^{-2}	1oo1	T4	2.8×10^{-2}	1oo1	T1	5.2×10^{-4}	1oo2	T4	1.1×10^{-6}	1oo2	T5
	EO vessels	2.0×10^{-3}	1oo2	T1	2.1×10^{-2}	1oo1	T1	5.8×10^{-4}	1oo2	T3	2.9×10^{-4}	1oo2	T5
	Ethylene vessels	5.0×10^{-3}	1oo1	T5	2.0×10^{-2}	1oo1	T2	5.7×10^{-4}	1oo2	T3	2.4×10^{-4}	1oo2	T5
	Reactor	1.4×10^{-2}	1oo1	T3	5.0×10^{-3}	1oo1	T5	6.8×10^{-7}	1oo3	T1, T3 & T4	2.8×10^{-6}	1oo3	T1, T2 & T4
	EO Absorber	7.0×10^{-3}	1oo1	T5	9.0×10^{-3}	1oo1	T4	4.2×10^{-7}	1oo3	T1, T3 & T4	1.7×10^{-6}	1oo3	T1, T2 & T4
F	CO_2 Absorber	1.1×10^{-2}	1oo1	T4	7.0×10^{-3}	1oo1	T5	6.7×10^{-7}	1oo3	T1, T3 & T4	1.1×10^{-6}	1oo3	T1, T3 & T5
	EO vessels	2.0×10^{-2}	1oo1	T1	4.0×10^{-3}	1oo1	T5	5.1×10^{-7}	1oo3	T1, T3 & T4	2.1×10^{-6}	1oo3	T1, T2 & T4
	Ethylene vessels	2.0×10^{-2}	1oo1	T1	3.0×10^{-3}	1oo1	T5	1.1×10^{-6}	1oo3	T1, T2 & T3	1.9×10^{-6}	1oo3	T1, T2 & T4

when a minimum tolerable risk is established. Also, it is remarkable that the quality of the solutions depends on the models used to quantify the consequences. The different models used into the consequences analysis can provide a different final result. Then, conservative or more practical consequences scenarios can be analyzed by the decision-makers using several models of effects.

It is important to mention that plant layout is an essential component of the overall operations of the company. In Mannan (2005), the book authors said: "additional space tends to increase safety, but is expensive in terms of land and also in additional pipework and operating costs. Space needs to be provided where it is necessary for safety, but not wasted". Then, the plant layout has a significant impact on plant economics, and it should enable to operate under the safety guidelines. If well the focus of this research has been related to the process safety, the economic aspect of the initial costs and the operating cost have been taking into account.

6. Conclusions

A MINLP approach has been proposed for solving the plant layout problem with the objective of reducing the risk of the process plant. The mathematical model minimizes the land, interconnection and SIS lifecycle costs subject to several constraints such as distances, non-overlapping, orientation, area, damage probabilities and the lifecycle of the SIS. The damage and escalated consequences were assessed using probit functions. The SIS design includes procurement, operating and risk costs. Also, the SIS probability of failure on demand was achieved taking into account specific architecture, multiple failure modes, diagnostic coverage, common cause failure, proof test intervals and repair times, fault tolerance, and the safe failure fraction.

The proposed model determines the optimal location of the equipment and facilities, and the design of the safety instrumented system in order to reduce possible accidents and the financial risk associated with individual units. The approach has successfully been applied to an ethylene oxide plant, thus demonstrating its applicability. The non-convexities presented do not guarantee a global optimization; however, the results provide valuable information for the decision makers. The optimal solution depends on the quality of the information provided and the models used to estimate the overpressure of the exposition. Moreover, non-SIS protection devices could be included to solve the case study by adding their cost and risk reduction factor (reducing the SIS cost). Nevertheless, the optimal layout solution was obtained only for the SIS installation in order to provide a better analysis of the applicability of the proposed model. In addition, the mathematical formulation can reduce the time and subjectivity in the analysis.

Finally, this research provides an excellent tool to obtain a preliminary solution to the plant layout. The proposed mathematical model application can help:

- to reduce future changes in the process equipment and facilities locations because the redesign is highly expensive,
- to improve the products, equipment, and containers handling,
- to optimize the space used,
- to minimize the pipework,
- to determine the appropriated localization of the buildings to limit the people exposure to hazards, administration buildings especially,
- to obtain appropriate segregation of the hazardous equipment in order to reduce the domino effect frequency and protect the vulnerable targets,
- to guarantee the safety of the process and workers that impact on productivity, and
- to effectively operate according to the legislation related to the safety guidelines.

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Appendix A

Fault Tree Analysis technique was selected to formulate *PFD* and *STR* equations. The systems analyzed have a 1 -out-of- n structure. However, k -out-of- n different structure can be included in the model solving the fault tree previously. The *PFD* estimation is assumed that the system operates in a low-demand mode, and the failure rate is constant. Redundancy allocation with diverse technology options is introduced as a defense against CCF (Common Cause Failures). Also, it has been assumed that appropriate measures have been put in place to eliminate coupling among different technologies; so that common cause failures of different technologies are negligible. Finally, this section presents the $C^{DPIC}_{i,s,tr}$, $C^{T}_{i,s,tr}$, $C^{PM}_{i,s,tr}$, and $C^{CM}_{i,s,tr}$, equations. The options taking into account for the study case were 1 -out-of- n structure with homogeneous technology redundancy, nA, and 1 -out-of- n structure with heterogeneous technology redundancy such as AB, $2AB$, $3AB$, $2A2B$, ABC, $2ABC$, and $ABCD$, where A, B, C, and D are four different technologies. Therefore, $rt \in TR = nA \cup 2AB \cup 3AB \cup AB \cup 2A2B \cup ABC \cup 2ABC \cup ABCD$. The sets and equations for each set are defined below.

- nA is the set of 1 -out-of- n structure, 1oo1 without redundancy and homogeneous redundancy structure for any technology available The equations for 1 -out-of- n without diverse technologies are:

$$\begin{aligned} PFD^{nA}_{i,s,t,r} = {} & \frac{\left[\lambda^{DU}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{DU}_{i,s,t}\right)(8760\cdot TI_i+Tr_i)+\lambda^{DD}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{DD}_{i,s,t}\right)Tr_i\right]^{N^{i,s,t,r}+1}}{8760\cdot TI_i\cdot\lambda^{DU}_{i,s,t}\left(N^{i,s,t,r}+1\right)\left(1-C^{MooN}_{i,r}\cdot\beta^{DU}_{i,s,t}\right)} \\ & -\frac{\left[\lambda^{DU}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{DU}_{i,s,t}\right)Tr_i+\lambda^{DD}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{DD}_{i,s,t}\right)Tr_i\right]^{N^{i,s,t,r}+1}}{8760\cdot TI_i\cdot\lambda^{DU}_{i,s,t}\left(N^{i,s,t,r}+1\right)\left(1-C^{MooN}_{i,r}\cdot\beta^{DU}_{i,s,t}\right)} \\ & +\lambda^{DU}_{i,s,t}\cdot C^{MooN}_{i,r}\cdot\beta^{DU}_{i,s,t}(4380\cdot TI_i+Tr_i)+\lambda^{DD}_{i,s,t}\cdot C^{MooN}_{i,r}\cdot\beta^{DD}_{i,s,t}\cdot Tr_i \forall i\in H;\ \forall s\in SUB; \\ & \forall t\in TEC;\ \forall r\in RED; \end{aligned} \tag{A.1}$$

where N is the number of redundancies with the same technology

$$\begin{aligned} STR^{nA}_{i,s,t} = {} & \left[\lambda^{SD}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{SD}_{i,s,t}\right)+\lambda^{SU}_{i,s,t}\left(1-C^{MooN}_{i,r}\cdot\beta^{SU}_{i,s,t}\right)\right]N^{i,s,t,r}+\lambda^{SD}_{i,s,t}\left(C^{MooN}_{i,r}\cdot\beta^{SD}_{i,s,t}\right) \\ & +\lambda^{SU}_{i,s,t}\left(C^{MooN}_{i,r}\cdot\beta^{SU}_{i,s,t}\right)\forall i\in H;\forall s\in SUB;\ \forall t\in TEC;\forall r\in RED \end{aligned} \tag{A.2}$$

$$CDPIC^{nA}_{i,s,t,r} = \left(C^{Design}_{i,s,t}+C^{Purchase}_{i,s,t}+C^{Installation}_{i,s,t}+C^{Commissioning}_{i,s,t}\right)N_{i,s,t,r}\quad \forall i\in H;\ \forall s\in SUB;\ \forall t\in TEC;\forall r\in RED; \tag{A.3}$$

$$C^{PM}_{i,s,t,r} = \sum_t\sum_r\frac{1}{M_t}C^{PM}_{i,s,t}N_{i,s,t,r}\quad \forall i\in H;\forall s\in SUB;\forall t\in TEC;\forall r\in RED \tag{A.4}$$

where M is the maintenance frequency.

$$C^{T}_{i,s,t,r} = \sum_t\sum_r\frac{1}{TI_i}C^{T}_{i,s,t}N^{i,s,t,r}\quad \forall i\in H;\forall s\in SUB;\forall t\in TEC;\forall r\in RED \tag{A.5}$$

where TI is the Test Interval

$$C^{CM}_{i,s,t,r} = 8760\sum_t\sum_r f_{i,s,t}C^{CM}_{i,s,t}N^{i,s,t,r}\quad \forall i\in H;\forall s\in SUB;\forall t\in TEC;\forall r\in RED \tag{A.6}$$

where $f_{i,s,t}=\lambda^{Total}_{i,s,t}$ is the total failure frequency of the device.

- AB is the set of 1oo2 structure with two different technologies. The equations are:

$$PFD^{AB}_{i,s,t,t2} = \left[\lambda^{DU}_{i,s,t}\cdot(4380\cdot TI_i+Tr_i)+\lambda^{DD}_{i,s,t}\cdot Tr_i\right]\cdot\left[\lambda^{DU}_{i,s,t2}\cdot(4380\cdot TI_i+Tr_i)+\lambda^{DD}_{i,s,t2}\cdot Tr_i\right]\quad \forall i\in H;\forall s\in SUB;\forall t,t2\in TEC;t2\neq t \tag{A.7}$$

$$STR^{AB}_{i,s,t,t2} = \lambda^{SD}_{i,s,t}+\lambda^{SU}_{i,s,t}+\lambda^{SD}_{i,s,t2}+\lambda^{SU}_{i,s,t2}\quad \forall i\in H;\forall s\in SUB;\forall t,t2\in TEC;t2\neq t; \tag{A.8}$$

$$\begin{aligned} CDPIC^{AB}_{i,s,t,t2} = {} & C^{Design}_{i,s,t}+C^{Purchase}_{i,s,t}+C^{Installation}_{i,s,t}+C^{Commissioning}_{i,s,t}+C^{Design}_{i,s,t2}+C^{Purchase}_{i,s,t2} \\ & +C^{Installation}_{i,s,t2}+C^{Commissioning}_{i,s,t2}\forall i\in H;\forall s\in SUB;\forall t,t2\in TEC;t2\neq t \end{aligned} \tag{A.9}$$

$$C^{PM,AB}_{i,s,t,t2} = \frac{1}{M_t}C^{PM}_{i,s,t}+\frac{1}{M_{t2}}C^{PM}_{i,s,t2}\quad \forall i\in H;\forall s\in SUB;\forall t,t2\in TEC;t2\neq t; \tag{A.10}$$

$$C^{T,AB}_{i,s,t,t2} = \frac{1}{TI_i}\left(C^{T}_{i,s,t}+C^{T}_{i,s,t2}\right)\quad \forall i\in H;\forall s\in SUB;\forall t,t2\in TEC;t2\neq t; \tag{A.11}$$

$$C_{i,s,t,t2}^{CM,AB} = 8760\left(f_{i,s,t}C_{i,s,t}^{CM} + f_{i,s,t2}C_{i,s,t2}^{CM}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.12}$$

- 2*AB* is the set of 1oo3 structure with two different technologies, 2 items with A-technology and one with B-technology. The equations are:

$$PFD_{i,s,t,t2,r}^{2AB} = \frac{\left[\lambda_{i,s,t}^{DU}\cdot(8760\cdot TI_i + Tr_i) + \lambda_{i,s,t}^{DD}\cdot Tr_i\right]^3 - \left[\left(\lambda_{i,s,t}^{DU} + \lambda_{i,s,t}^{DD}\right)\cdot Tr_i\right]^3}{3\cdot 8760\cdot TI_i\cdot\lambda_{i,s,t}^{DU}} \cdot\lambda_{i,s,t2}^{DU}\cdot(4380\cdot TI_i + Tr_i) + \lambda_{i,s,t2}^{DD}\cdot Tr_i \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.13}$$

$$STR_i^{2AB} = 2\left[\lambda_{i,s,t}^{SD}\left(1 - C_{i,r}^{Moo2}\cdot\beta_{i,s,t}^{SD}\right) + \lambda_{i,s,t}^{SU}\left(1 - C_{i,r}^{Moo2}\cdot\beta_{i,s,t}^{SU}\right)\right] + \lambda_{i,s,t2}^{SD} + \lambda_{i,s,t2}^{SU} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.14}$$

$$CDPIC_{i,s,t,t2,r}^{2AB} = 2\left(C_{i,s,t}^{Design} + C_{i,s,t}^{Purchase} + C_{i,s,t}^{Installation} + C_{i,s,t}^{Commissioning}\right) + C_{i,s,t2}^{Design} + C_{i,s,t2}^{Purchase} + C_{i,s,t2}^{Installation} + C_{i,s,t2}^{Commissioning} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.15}$$

$$C_{i,s,t,t2,r}^{PM,2AB} = \frac{2}{M_t}C_{i,s,t}^{PM} + \frac{1}{M_{t2}}C_{i,s,t2}^{PM} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.16}$$

$$C_{i,s,t,t2,r}^{T,2AB} = \frac{1}{TI_i}\left(2C_{i,s,t}^{T} + C_{i,s,t2}^{T}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.17}$$

$$C_{i,s,t,r}^{CM,2AB} = 8760\left(2f_{i,s,t}C_{i,s,t}^{CM} + f_{i,s,t2}C_{i,s,t2}^{CM}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.18}$$

- 3*AB* is the set of 1oo4 structure with two different technologies, 3 items with A-technology and one with B-technology. The equations are:

$$PFD_{i,s,t,t2,r}^{3AB} = \frac{\left[\lambda_{i,s,t}^{DU}\cdot(8760\cdot TI_i + Tr_i) + \lambda_{i,s,t}^{DD}\cdot Tr_i\right]^4 - \left[\left(\lambda_{i,s,t}^{DU} + \lambda_{i,s,t}^{DD}\right)\cdot Tr_i\right]^4}{4\cdot 8760\cdot TI_i\cdot\lambda_{i,s,t}^{DU}}\cdot\lambda_{i,s,t2}^{DU} \cdot(4380\cdot TI_i + Tr_i) + \lambda_{i,s,t2}^{DD}\cdot Tr_i \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.19}$$

$$STR_i^{3AB} = 3\left[\lambda_{i,s,t}^{SD}\left(1 - C_{i,r}^{Moo2}\cdot\beta_{i,s,t}^{SD}\right) + \lambda_{i,s,t}^{SU}\left(1 - C_{i,r}^{Moo2}\cdot\beta_{i,s,t}^{SU}\right)\right] + \lambda_{i,s,t2}^{SD} + \lambda_{i,s,t2}^{SU} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.20}$$

$$CDPIC_{i,s,t,t2,r}^{3AB} = 3\left(C_{i,s,t}^{Design} + C_{i,s,t}^{Purchase} + C_{i,s,t}^{Installation} + C_{i,s,t}^{Commissioning}\right) + C_{i,s,t2}^{Design} + C_{i,s,t2}^{Purchase} + C_{i,s,t2}^{Installation} + C_{i,s,t2}^{Commissioning} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.21}$$

$$C_{i,s,t,t2,r}^{PM,3AB} = \frac{3}{M_t}C_{i,s,t}^{PM} + \frac{1}{M_{t2}}C_{i,s,t2}^{PM} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.22}$$

$$C_{i,s,t,t2,r}^{T,3AB} = \frac{1}{TI_i}\left(3C_{i,s,t}^{T} + C_{i,s,t2}^{T}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.23}$$

$$C_{i,s,t,r}^{CM,3AB} = 8760\left(3f_{i,s,t}C_{i,s,t}^{CM} + f_{i,s,t2}C_{i,s,t2}^{CM}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \forall r \in RED \tag{A.24}$$

- 2*A*2*B* is the set of 1oo4 structure with two different technologies, 2 items with A-technology and 2 items with B-technology. The equations are:

$$PFD_{i,s,t,t2}^{2A2B} = \frac{1}{9\cdot\lambda_{i,s,t}^{DU}\cdot\lambda_{i,s,t2}^{DU}\cdot TI_i^2\cdot 8760^2}\cdot\left\{\left[\lambda_{i,s,t}^{DU}\cdot(8760\cdot TI_i + Tr_i) + \lambda_{p,s,t}^{DD}\cdot Tr_i\right]^3 - \left[\left(\lambda_{i,s,t}^{DU} + \lambda_{i,s,t}^{DU}\right)Tr_i\right]^3\right\} \cdot\left\{\left[\lambda_{i,s,t2}^{DU}\cdot(8760\cdot TI_i + Tr_i) + \lambda_{i,s,t2}^{DD}\cdot Tr_i\right]^3 - \left[\left(\lambda_{i,s,t2}^{DU} + \lambda_{p,s,t2}^{DU}\right)Tr_i\right]^3\right\} \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.25}$$

$$STR_{i,s,t,t2}^{2A2B} = 2\left(\lambda_{i,s,t}^{SD} + \lambda_{i,s,t}^{SU} + \lambda_{i,s,t2}^{SD} + \lambda_{i,s,t2}^{SU}\right) \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.26}$$

$$CDPIC_{i,s,t,t2}^{2A2B} = 2\cdot CDPIC_{i,s,t,t2}^{AB} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.27}$$

$$C_{i,s,t,t2}^{PM,2A2B} = 2\cdot C_{i,s,t,t2}^{PM,AB} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.28}$$

$$C_{i,s,t,t2}^{T,2A2B} = 2\cdot C_{i,s,t,t2}^{T,AB} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.29}$$

$$C_{i,s,t}^{CM,2A2B} = 2\cdot C_{i,s,t2}^{CM,AB} \quad \forall i \in H; \forall s \in SUB; \forall t, t2 \in TEC; t2 \neq t; \tag{A.30}$$

- *ABC* is the set of 1oo3 structure with three different technologies. The equations are:

$$PFD^{ABC}_{i,s,t,t2,t3} = \left[\lambda^{DU}_{i,s,t}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t}\cdot Tr_i\right]\cdot\left[\lambda^{DU}_{i,s,t2}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t2}\cdot Tr_i\right] \cdot\left[\lambda^{DU}_{i,s,t3}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t3}\cdot Tr_i\right]\ \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.31)$$

$$STR^{ABC}_{i,s,t,t2,t3} = \lambda^{SD}_{i,s,t} + \lambda^{SU}_{i,s,t} + \lambda^{SD}_{i,s,t2} + \lambda^{SU}_{i,s,t2} + \lambda^{SD}_{i,s,t3} + \lambda^{SU}_{i,s,t3} \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.32)$$

$$CDPIC^{ABC}_{i,s,t,t2,t3} = C^{Design}_{i,s,t} + C^{Purchase}_{i,s,t} + C^{Installation}_{i,s,t} + C^{Commissioning}_{i,s,t} + C^{Design}_{i,s,t2} + C^{Purchase}_{i,s,t2} + C^{Installation}_{i,s,t2} + C^{Commissioning}_{i,s,t2} + C^{Design}_{i,s,t3} + C^{Purchase}_{i,s,t3} + C^{Installation}_{i,s,t3} + C^{Commissioning}_{i,s,t3} \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.33)$$

$$C^{PM,ABC}_{i,s,t,t2,t3} = \frac{1}{M_t}C^{PM}_{i,s,t} + \frac{1}{M_{t2}}C^{PM}_{i,s,t2} + \frac{1}{M_{t3}}C^{PM}_{i,s,t3} \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.34)$$

$$C^{T,ABC}_{i,s,t,t2,t3} = \frac{1}{TI_i}\left(C^{T}_{i,s,t} + C^{T}_{i,s,t2} + C^{T}_{i,s,t3}\right) \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.35)$$

$$C^{CM,ABC}_{i,s,t,t2,t3} = 8760\left(f_{i,s,t}C^{CM}_{i,s,t} + f_{i,s,t2}C^{CM}_{i,s,t2} + f_{i,s,t3}C^{CM}_{i,s,t3}\right) \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.36)$$

- 2*ABC* is the set of 1oo4 structure with three different technologies. 2 items with A-technology, one item with B-technology and one item with C-technology. The equations are:

$$PFD^{2ABC}_{i,s,t,t2,t3} = \frac{1}{3\cdot\lambda^{DU}_{i,s,t}\cdot TI^{i}\cdot 8760}\left\{\left[\lambda^{DU}_{i,s,t}\cdot(8760\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t}\cdot Tr_i\right]^3 - \left[\left(\lambda^{DU}_{i,s,t} + \lambda^{DU}_{i,s,t}\right)Tr_i\right]^3\right\} \cdot\left[\lambda^{DU}_{i,s,t2}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t2}\cdot Tr_i\right]\cdot\left[\lambda^{DU}_{i,s,t3}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t3}\cdot Tr_i\right]\forall i \in H;\forall s \in SUB;\forall t, t2 \in TEC; t2 \neq t; \quad (A.37)$$

$$STR^{2ABC}_{i,s,t,t2,t3} = 2\left(\lambda^{SD}_{i,s,t} + \lambda^{SU}_{i,s,t}\right) + \lambda^{SD}_{i,s,t2} + \lambda^{SU}_{i,s,t2} + \lambda^{SD}_{i,s,t3} + \lambda^{SU}_{i,s,t3} \quad \forall \in \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3 \quad (A.38)$$

$$CDPIC^{2ABC}_{i,s,t,t2,t3} = 2\left(C^{Design}_{i,s,t} + C^{Purchase}_{i,s,t} + C^{Installation}_{i,s,t} + C^{Commissioning}_{i,s,t}\right) + C^{Design}_{i,s,t2} + C^{Purchase}_{i,s,t2} + C^{Installation}_{i,s,t2} + C^{Commissioning}_{i,s,t2} + C^{Design}_{i,s,t3} + C^{Purchase}_{i,s,t3} + C^{Installation}_{i,s,t3} + C^{Commissioning}_{i,s,t3} \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3; \quad (A39)$$

$$C^{PM,2ABC}_{i,s,t,t2,t3} = \frac{2}{M_t}C^{PM}_{i,s,t} + \frac{1}{M_{t2}}C^{PM}_{i,s,t2} + \frac{1}{M_{t3}}C^{PM}_{i,s,t3} \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3; \quad (A.40)$$

$$C^{T,2ABC}_{i,s,t,t2,t3} = \frac{2}{TI_i}\left(C^{T}_{i,s,t} + C^{T}_{i,s,t2} + C^{T}_{i,s,t3}\right) \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3; \quad (A.41)$$

$$C^{CM,2ABC}_{i,s,t,t2,t3} = 8760\left(2f_{i,s,t}C^{CM}_{p,s,t} + f_{i,s,t2}C^{CM}_{i,s,t2} + f_{i,s,t3}C^{CM}_{i,s,t3}\right) \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3 \in TEC; t \neq t2 \neq t3; \quad (A42)$$

- *ABCD* is the set of 1oo4 structure with four different technologies. The equations are:

$$PFD^{ABCD}_{i,s,t,t2,t3,t4} = \left[\lambda^{DU}_{i,s,t}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t}\cdot Tr_i\right]\cdot\left[\lambda^{DU}_{i,s,t2}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{i,s,t2}\cdot Tr_i\right] \cdot\left[\lambda^{DU}_{p,s,t3}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{p,s,t3}\cdot Tr_i\right]\cdot\left[\lambda^{DU}_{p,s,t4}\cdot(4380\cdot TI_i + Tr_i) + \lambda^{DD}_{p,s,t4}\cdot Tr_i\right] \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.43)$$

$$STR^{ABCD}_{i,s,t,t2,t3,t4} = \lambda^{SD}_{i,s,t} + \lambda^{SU}_{i,s,t} + \lambda^{SD}_{i,s,t2} + \lambda^{SU}_{i,s,t2} + \lambda^{SD}_{i,s,t3} + \lambda^{SU}_{i,s,t3} + \lambda^{SD}_{i,s,t4} + \lambda^{SU}_{i,s,t4} \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.44)$$

$$CDPIC^{ABCD}_{i,s,t,t2,t3.t4} = C^{Design}_{i,s,t} + C^{Purchase}_{i,s,t} + C^{Installation}_{i,s,t} + C^{Commissioning}_{i,s,t} + C^{Design}_{i,s,t2} + C^{Purchase}_{i,s,t2} + C^{Installation}_{i,s,t2} + C^{Commissioning}_{i,s,t2} + C^{Design}_{i,s,t3} + C^{Purchase}_{i,s,t3} + C^{Installation}_{i,s,t3} + C^{Commissioning}_{i,s,t3} + C^{Design}_{i,s,t4} + C^{Purchase}_{i,s,t4} + C^{Installation}_{i,s,t4} + C^{Commissioning}_{i,s,t4} \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.45)$$

$$C^{PM,ABCD}_{i,s,t,t2,t3,t4} = \frac{1}{M_t}C^{PM}_{i,s,t} + \frac{1}{M_{t2}}C^{PM}_{i,s,t2} + \frac{1}{M_{t3}}C^{PM}_{i,s,t3} + \frac{1}{M_{t4}}C^{PM}_{i,s,t4} \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.46)$$

$$C^{T,ABCD}_{i,s,t,t2,t3,t4} = \frac{1}{TI_i}\left(C^{T}_{i,s,t} + C^{T}_{i,s,t2} + C^{T}_{i,s,t3} + C^{T}_{i,s,t4}\right) \quad \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.47)$$

$$C^{CM,ABCD}_{i,s,t,t2,t3,t4} = 8760\left(f_{i,s,t}C^{CM}_{i,s,t} + f_{i,s,t2}C^{CM}_{i,s,t2} + f_{i,s,t3}C^{CM}_{i,s,t3} + f_{i,s,t4}C^{CM}_{i,s,t4}\right) \forall i \in H;\forall s \in SUB;\forall t, t2, t3, t4 \in TEC; t \neq t2 \neq t3 \neq t4; \quad (A.48)$$

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